

Sample machine documentation

Pressing and marking line LL-03 (complex example)

Parts feeder, vacuum manipulator, hydraulic press with marking and outfeed conveyor; two-hand start, light curtain, two guards.

29 devices · 44 I/O signals · generated 03/10/2026

APPROVAL STATUS: 40 OF 73 ITEMS APPROVED, 3 CHANGED AFTER APPROVAL. THE REST ARE MARKED NOT APPROVED.

Sample. The approving person is fictitious. Generated in PLCdesk from the sample project without manual edits.

01 Overview

documentation contents and checklist of missing documents

02 Approval

approval of design items: status, by whom, when, note

03 FDS

functional design specification

04 I/O list

complete I/O list

05 Terminal strip

terminal strip for the electrical designer

06 Block diagram

00_blokove_schema.svg

07 Wiring of input module DI1

01_DI1_X1.svg

08 Safety circuit schematic

00_bezpecnostni_okruh.svg

09 Alarms

alarm list with reactions and acknowledgement

10 FAT

test protocol (loop check, functional tests, sequence)

11 Manual

operating manual skeleton

12 Safety

safety functions: hazards, PLr, architecture, PL calculation, wiring (SRS)

13 Safety program

safety_00_CEGA_NA_SCHVALENI.md

14 Commissioning

commissioning plan and report by phase

15 Program for Siemens (excerpt)

siemens_Gen_Main.scl

Project documentation overview

CHANGED AFTER APPROVAL — content changed after approval: Automatic sequence, Cycle time target and watchdog times, Simulation verification result. The approval is no longer valid — approve again (see 11_schvaleni.md).

Project: Pressing and marking line LL-03 (complex example) · generated on 03/10/2026 by PLCdesk

Contents

- 00_prehled_dokumentace.md — documentation contents and checklist of missing documents
- 01_funkcni_specifikace_FDS.md — functional design specification
- 02_io_list.csv — complete I/O list
- 03_svorkovnice.csv — terminal strip for the electrical designer
- 04_seznam_alarmu.csv — alarm list with reactions and acknowledgement
- 05_testovaci_protokol_FAT.md — test protocol (loop check, functional tests, sequence)
- 06_navod_k_obsluze.md — operating manual skeleton
- 07_softwarova_dokumentace.md — program structure and conventions
- 08_overeni_simulaci.md — verification of the sequence by process simulation (normal cycle, faults, E-stop)
- 09_kusovnik.md — component bill of materials with brands and suppliers
- 09_kusovnik.csv — bill of materials for Excel / request for quotation
- 11_schvaleni.md — approval of design items: status, by whom, when, note
- 12_protokol_ozivovani.md — commissioning plan and report by phase
- 12_protokol_ozivovani.csv — commissioning report for printing and filling in
- diagrams — block diagram, function chart and timing diagram of the cycle, I/O wiring diagram (SVG preview + DXF for CAD)
- program source files for each selected platform including the import procedure (README)

Documents that the set does NOT contain and a typical project requires (add manually)

- URS / FRS (customer requirements and functional requirements)
- Risk assessment and validation of safety functions (ISO 13849 / IEC 62061)
- Bill of materials (BOM) and control cabinet drawings, as-built electrical documentation
- Network topology and IP plan, HMI/SCADA specification
- SAT protocol, IQ/OQ (depending on the industry), program backups, declaration of conformity (CE)

The generated documents are an initial draft for review — check and complete them before handing them over to the customer.

Project approval

CHANGED AFTER APPROVAL — content changed after approval: Automatic sequence, Cycle time target and watchdog times, Simulation verification result. The approval is no longer valid — approve again (see 11_schvaleni.md).

Project: Pressing and marking line LL-03 (complex example)

PLCdesk proposes, a responsible person approves. Each item is approved with a name, date and note, and the approval applies to the content at the moment of approval (fingerprint). When the content of an item changes, its status becomes “changed after approval” and the item must be approved again. Nothing is approved automatically.

Status: 40 approved · 3 changed after approval · 0 rejected · 30 pending — 73 items in total

Mandatory items without valid approval (17)

- Automatic sequence — changed after approval
- Cycle time target and watchdog times — changed after approval
- Simulation verification result — changed after approval
- Commissioning plan — not approved
- SF5 — Light curtain (AOPD) – danger point protection (finger/hand) — S4 — not approved
- SF5 — architecture, components and PL calculation — not approved
- SF6 — Two-hand control — S5, S6 — not approved
- SF6 — architecture, components and PL calculation — not approved
- SF7 — Prevention of unexpected start-up (isolation and energy dissipation, LOTO, start interlock) — not approved
- SF7 — architecture, components and PL calculation — not approved
- SF8 — Safe pneumatic exhaust (SDE) and blocking of pneumatic movements (SSC) — not approved
- SF8 — architecture, components and PL calculation — not approved
- SF9 — Safe stop / isolation of hydraulics (redundant valves with position monitoring) — not approved
- SF9 — architecture, components and PL calculation — not approved
- SF10 — Pressure limitation (relief valve) and pressure monitoring — not approved
- SF10 — architecture, components and PL calculation — not approved
- Safety program (Siemens F-program) — not approved

1. Device and I/O design

Item	What is approved	Mandatory	Status	By	When	Note	Fingerprint
Device M1 (Infeed belt conveyor)	Motor / pump; running feedback, fault input	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	63686682
Device M2 (Outfeed belt conveyor)	Motor / pump; running feedback, fault input	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	c315b201
Device M3 (Vibratory parts feeder)	Motor / pump	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	b5b471e5
Device M4 (Hydraulic power unit pump)	Motor / pump; running feedback, fault input	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	81750244
Device Y1 (Press cylinder)	Valve / cylinder; open limit switch, closed limit switch	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	0cf8d122
Device Y2 (Left clamping cylinder)	Valve / cylinder; open limit switch, closed limit switch	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	839a40e6
Device Y3 (Right clamping cylinder)	Valve / cylinder; open limit switch, closed limit switch	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	e7d936dd
Device Y4 (Manipulator — transfer into the press)	Valve / cylinder; open limit switch, closed limit switch	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	e90dffdd
Device Y5 (Manipulator vacuum suction cup)	Valve / cylinder; open limit switch	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	4e82c216
Device Y6 (Marking unit (stamping cylinder))	Valve / cylinder	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	c47cd2f1
Device B1 (Hydraulic pressure)	Analog input; range 0–250 bar; no limits	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	5897a531
Device B2 (Oil temperature)	Analog input; range 0–100 °C; no limits	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	bdc9286a

Item	What is approved	Mandatory	Status	By	When	Note	Fingerprint
Device B3 (Press force (strain gauge))	Analog input; range 0–400 kN; no limits	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	7677b4d
Device B4 (Press position (linear sensor))	Analog input; range 0–500 mm; no limits	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	87b26a00
Device B5 (Cooling flow)	Analog input; range 0–60 l/min; no limits	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	9ef05399
Device U1 (Proportional pressure valve)	Analog output; range 0–100 %; no setpoint	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	f87db4eb
Device U2 (Conveyor M2 speed setpoint)	Analog output; range 0–100 %; no setpoint	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	f1d0197e
Device S1 (Emergency stop (NC))	Digital input (sensor)	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	d45b7db6
Device S2 (Left guard closed (NC))	Digital input (sensor)	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	cc8b58b6
Device S3 (Right guard closed (NC))	Digital input (sensor)	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	c3f02fb6
Device S4 (Light curtain clear)	Digital input (sensor)	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	ffb84ab7
Device S5 (Two-hand start — button A)	Digital input (sensor)	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	f7db21b7
Device S6 (Two-hand start — button B)	Digital input (sensor)	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	ef31fcb7
Device S7 (Part in position under the manipulator)	Digital input (sensor)	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	e5bbd3b7
Device S8 (Parts magazine empty)	Digital input (sensor)	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	98a05eb6
Device H1 (Beacon green — running)	Digital output (indicator lamp...); not linked to the machine state	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	3fb554cf
Device H2 (Beacon red — fault)	Digital output (indicator lamp...); not linked to the machine state	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	98dd592e
Device H3 (Horn — cycle start)	Digital output (indicator lamp...); not linked to the machine state	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	93ecf15b
Device H4 (Electromagnetic guard lock)	Digital output (indicator lamp...); not linked to the machine state	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	adcce9be
I/O table	44 signals (DI 23, DO 14, AI 5, AO 2); tags, addresses, normally closed contacts	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	9c958f35

2. Program

Item	What is approved	Mandatory	Status	By	When	Note	Fingerprint
Automatic sequence	18 steps: order, actions, transitions and their times; AUTO / manual modes	yes	changed after approval	Jan Sample	02/10/2026, 11:00	sample, fictitious person	18896531
E-stop and interlocks (standard program signals)	E-stop: S1; interlocks: S2, S3, S4. Central enable inputs of the standard program — safety functions do not belong here.	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	be354364
Cycle time target and watchdog times	cycle time target not set; 13 step watchdog times; block timeouts: motor 3 s, valve 5 s	yes	changed after approval	Jan Sample	02/10/2026, 11:00	sample, fictitious person	98921616

3. Simulation verification

Item	What is approved	Mandatory	Status	By	When	Note	Fingerprint
Simulation verification result	no errors; 14 warnings; cycle 23.31 s; state matrix 159/159	yes	changed after approval	Jan Sample	02/10/2026, 11:00	sample, fictitious person	3607ca57

4. Safety

Item	What is approved	Mandatory	Status	By	When	Note	Fingerprint
Hazards and assignment of safety functions	8 hazards, 10 proposed functions (0 excluded); hazard identification and choice of measures (ISO 12100) — the application does not decide that the risk is acceptable.	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	fe952649
SF1 — Emergency stop (E-stop) — S1	PLr d (S2/F1/P2); stop category 0; triggered by: S1	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	1c5f9dcd
SF1 — architecture, components and PL calculation	cat. 3, 2 channel(s), EDM yes; achieved PL d ≥ PLr d; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	abc06db9
SF2 — Interlocking of a movable guard (without guard locking) — S2	PLr e (S2/F2/P2); stop category 0; triggered by: S2	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	4e942670
SF2 — architecture, components and PL calculation	cat. 4, 2 channel(s), EDM yes; achieved PL e ≥ PLr e; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	3543881c
SF3 — Interlocking of a movable guard (without guard locking) — S3	PLr e (S2/F2/P2); stop category 0; triggered by: S3	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	d2f8259e
SF3 — architecture, components and PL calculation	cat. 4, 2 channel(s), EDM yes; achieved PL e ≥ PLr e; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	0228eb9f
SF4 — Guard with guard locking and release of the lock after stopping — S2, S3, H4	PLr d (S2/F1/P2); stop category 0; triggered by: S2, S3, H4	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	2dd89587
SF4 — architecture, components and PL calculation	cat. 3, 2 channel(s), EDM yes; achieved PL d ≥ PLr d; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC	yes	approved	Jan Sample	02/10/2026, 11:00	sample, fictitious person	f7148cbb
SF5 — Light curtain (AOPD) – danger point protection (finger/hand) — S4	PLr e (S2/F2/P2); stop category 0; triggered by: S4	yes	not approved				6aa132c4
SF5 — architecture, components and PL calculation	cat. 4, 2 channel(s), EDM yes; achieved PL — < PLr e; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC; 1 open items	yes	not approved				ee791790
SF6 — Two-hand control — S5, S6	PLr e (S2/F2/P2); stop category 0; triggered by: S5, S6	yes	not approved				43b5eab9
SF6 — architecture, components and PL calculation	cat. 4, 2 channel(s), EDM yes; achieved PL e ≥ PLr e; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC	yes	not approved				bf9531ba
SF7 — Prevention of unexpected start-up (isolation and energy dissipation, LOTO, start interlock)	PLr d (S2/F1/P2); stop category 0; triggered by: —	yes	not approved				b66a7592
SF7 — architecture, components and PL calculation	cat. 3, 2 channel(s), EDM yes; achieved PL d ≥ PLr d; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC	yes	not approved				f974015a
SF8 — Safe pneumatic exhaust (SDE) and blocking of pneumatic movements (SSC)	PLr e (S2/F2/P2); stop category 0; triggered by: —	yes	not approved				dfcd5b3e
SF8 — architecture, components and PL calculation	cat. 4, 2 channel(s), EDM yes; achieved PL e ≥ PLr e; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC	yes	not approved				0ef2169e
SF9 — Safe stop / isolation of hydraulics (redundant valves with position monitoring)	PLr e (S2/F2/P2); stop category 0; triggered by: —	yes	not approved				82d08b19
SF9 — architecture, components and PL calculation	cat. 4, 2 channel(s), EDM yes; achieved PL e ≥ PLr e; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC	yes	not approved				9096aa93
SF10 — Pressure limitation (relief valve) and pressure monitoring	no PLr (mechanical principle); stop category 0	yes	not approved				0719575c
SF10 — architecture, components and PL calculation	cat. B, 1 channel(s), EDM yes; achieved PL — < PLr —; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC; 2 open items	yes	not approved				8c706fea

Item	What is approved	Mandatory	Status	By	When	Note	Fingerprint
Safety program (Siemens F-program)	safety program / configuration for SIMATIC S7-1200F CPU 1214FC DC/DC/DC: certified blocks for 9 functions, outputs KS, YS, YH; generated only after the functions are approved	yes	not approved				451f6fbf

5. Commissioning

Item	What is approved	Mandatory	Status	By	When	Note	Fingerprint
Commissioning plan	221 steps in 10 phases: control cabinet, loop test of every I/O signal, drives, analogs, E-stop and interlocks, manual mode, sequence, fault states, cycle time, safety validation	yes	not approved				10040f25
Commissioning closure	0 OK, 0 N/A, 0 NOK, 221 without result out of 221 steps	no	not approved				97a7a8ce

6. Tuning proposals

Item	What is approved	Mandatory	Status	By	When	Note	Fingerprint
Measurement B1 (Hydraulic pressure): add limits 12.5 / 237.5 bar	The program does not react to the value (no alarm or condition). Proposal from the range 0–250 bar: 5 % from the ends — a faulty sensor or a broken wire then raises a fault. Adjust to the process (operating limits are usually narrower).	no	not approved				3cfc6d35
Measurement B2 (Oil temperature): add limits 5 / 95 °C	The program does not react to the value (no alarm or condition). Proposal from the range 0–100 °C: 5 % from the ends — a faulty sensor or a broken wire then raises a fault. Adjust to the process (operating limits are usually narrower).	no	not approved				6a6ec3cc
Measurement B3 (Press force (strain gauge)): add limits 20 / 380 kN	The program does not react to the value (no alarm or condition). Proposal from the range 0–400 kN: 5 % from the ends — a faulty sensor or a broken wire then raises a fault. Adjust to the process (operating limits are usually narrower).	no	not approved				af227ca3
Measurement B4 (Press position (linear sensor)): add limits 25 / 475 mm	The program does not react to the value (no alarm or condition). Proposal from the range 0–500 mm: 5 % from the ends — a faulty sensor or a broken wire then raises a fault. Adjust to the process (operating limits are usually narrower).	no	not approved				3d502019
Measurement B5 (Cooling flow): add limits 3 / 57 l/min	The program does not react to the value (no alarm or condition). Proposal from the range 0–60 l/min: 5 % from the ends — a faulty sensor or a broken wire then raises a fault. Adjust to the process (operating limits are usually narrower).	no	not approved				2521d0a7
Setpoint U1 (Proportional pressure valve) is not specified	Without a setpoint the output stays at 0. Enter a constant, or add the source (HMI, recipe, control loop) manually.	no	not approved				71b93837
Setpoint U2 (Conveyor M2 speed setpoint) is not specified	Without a setpoint the output stays at 0. Enter a constant, or add the source (HMI, recipe, control loop) manually.	no	not approved				1c71f950
Input S5 (Two-hand start — button A) is not read by the program	Decide its function: interlock input (guard, light curtain, air pressure — add to interlocks), step condition (wait for input), or custom logic. Otherwise remove it from the design.	no	not approved				90f03584
Input S6 (Two-hand start — button B) is not read by the program	Decide its function: interlock input (guard, light curtain, air pressure — add to interlocks), step condition (wait for input), or custom logic. Otherwise remove it from the design.	no	not approved				85555128
Input S7 (Part in position under the manipulator) is not read by the program	Decide its function: interlock input (guard, light curtain, air pressure — add to interlocks), step condition (wait for input), or custom logic. Otherwise remove it from the design.	no	not approved				96c9d9d1
Input S8 (Parts magazine empty) is not read by the program	Decide its function: interlock input (guard, light curtain, air pressure — add to interlocks), step condition (wait for input), or custom logic. Otherwise remove it from the design.	no	not approved				4b15d43c
Output H1 (Beacon green — running) is not controlled by the program	The output stays permanently FALSE. Propose a link to the machine state (running, fault, ready, stop, guard lock, AUTO) or custom logic.	no	not approved				a81d920f
Output H2 (Beacon red — fault) is not controlled by the program	The output stays permanently FALSE. Propose a link to the machine state (running, fault, ready, stop, guard lock, AUTO) or custom logic.	no	not approved				a7b1bc0b
Output H3 (Horn — cycle start) is not controlled by the program	The output stays permanently FALSE. Propose a link to the machine state (running, fault, ready, stop, guard lock, AUTO) or custom logic.	no	not approved				f0d76cdd
Output H4 (Electromagnetic guard lock) is not controlled by the program	The output stays permanently FALSE. Propose a link to the machine state (running, fault, ready, stop, guard lock, AUTO) or custom logic.	no	not approved				3495ee2b

Fingerprint = checksum of the item content (FNV-1a over canonical JSON). It does not include descriptions and comments — editing them does not invalidate an approval.

Functional design specification (FDS)

CHANGED AFTER APPROVAL — content changed after approval: Automatic sequence, Cycle time target and watchdog times. The approval is no longer valid — approve again (see 11_schvaleni.md).

Project	Pressing and marking line LL-03 (complex example)
Date	03/10/2026
Revision	0.1 — draft (PLCdesk)

1. Machine description and purpose

Parts feeder, vacuum manipulator, hydraulic press with marking and outfeed conveyor; two-hand start, light curtain, two guards.

2. Target control systems

- Siemens SIMATIC — TIA Portal V17–V21, S7-1200 / S7-1500, language SCL

3. Devices

Designation	Class	Description	Parameters
M1	Motor / pump	Infeed belt conveyor	running feedback, fault input
M2	Motor / pump	Outfeed belt conveyor	running feedback, fault input
M3	Motor / pump	Vibratory parts feeder	—
M4	Motor / pump	Hydraulic power unit pump	running feedback, fault input
Y1	Valve / cylinder	Press cylinder	open limit switch, closed limit switch
Y2	Valve / cylinder	Left clamping cylinder	open limit switch, closed limit switch
Y3	Valve / cylinder	Right clamping cylinder	open limit switch, closed limit switch
Y4	Valve / cylinder	Manipulator — transfer into the press	open limit switch, closed limit switch
Y5	Valve / cylinder	Manipulator vacuum suction cup	open limit switch
Y6	Valve / cylinder	Marking unit (stamping cylinder)	—
B1	Analog input	Hydraulic pressure	bar 0–250
B2	Analog input	Oil temperature	°C 0–100
B3	Analog input	Press force (strain gauge)	kN 0–400
B4	Analog input	Press position (linear sensor)	mm 0–500
B5	Analog input	Cooling flow	l/min 0–60
U1	Analog output	Proportional pressure valve	% 0–100
U2	Analog output	Conveyor M2 speed setpoint	% 0–100
S1	Digital input (sensor)	Emergency stop (NC)	—
S2	Digital input (sensor)	Left guard closed (NC)	—
S3	Digital input (sensor)	Right guard closed (NC)	—
S4	Digital input (sensor)	Light curtain clear	—
S5	Digital input (sensor)	Two-hand start — button A	—
S6	Digital input (sensor)	Two-hand start — button B	—
S7	Digital input (sensor)	Part in position under the manipulator	—
S8	Digital input (sensor)	Parts magazine empty	—
H1	Digital output (indicator lamp...)	Beacon green — running	—
H2	Digital output (indicator lamp...)	Beacon red — fault	—
H3	Digital output (indicator lamp...)	Horn — cycle start	—
H4	Digital output (indicator lamp...)	Electromagnetic guard lock	—

4. I/O count and modules

DI: 23 · DO: 14 · AI: 5 · AO: 2 Proposed modules: DI1 (terminal strip X1, 16 channels), DI2 (terminal strip X2, 7 channels), DO1 (terminal strip X3, 14 channels), AI1 (terminal strip X4, 5 channels), AO1 (terminal strip X5, 2 channels)

5. Modes and operation

- **MANUAL** (`modeAuto` = FALSE) — commands to individual devices from the HMI: `manRun_<motor>` , `manOpen_<valve>` (HMI specification: to be completed)
- **AUTO** (`modeAuto` = TRUE) — automatic sequence as per section 6; start (`cmdAutoStart`) is conditional on the central enable and a fault-free state; manual commands have no effect in AUTO
- **Acknowledge** (`cmdAck`) — clears the machine fault and the block faults once the cause has been removed
- **Central enable** (`enable`) — S1 – Emergency stop (NC) AND interlock inputs: S2 – Left guard closed (NC), S3 – Right guard closed (NC), S4 – Light curtain clear. FALSE on any of them stops the machine: blocks switch off their outputs, the sequence returns to step 0; a new start is required after restoring

6. Automatic sequence

1. M3 (Vibratory parts feeder) — start · transition: time 4 s
2. M1 (Infeed belt conveyor) — start · transition: feedback (watchdog time 3 s)
3. Y4 (Manipulator — transfer into the press) — open · transition: feedback (watchdog time 5 s)
4. Y5 (Manipulator vacuum suction cup) — open · transition: feedback (watchdog time 2 s)
5. Y4 (Manipulator — transfer into the press) — close · transition: feedback (watchdog time 5 s)
6. Y2 (Left clamping cylinder) — open · transition: feedback (watchdog time 4 s)
7. Y3 (Right clamping cylinder) — open · transition: feedback (watchdog time 4 s)
8. M4 (Hydraulic power unit pump) — start · transition: feedback (watchdog time 3 s)
9. Y1 (Press cylinder) — open · transition: feedback (watchdog time 8 s)
10. dwell 5 s · transition: time 5 s
11. Y1 (Press cylinder) — close · transition: feedback (watchdog time 8 s)
12. M4 (Hydraulic power unit pump) — stop · transition: feedback (watchdog time 3 s)
13. Y6 (Marking unit (stamping cylinder)) — open · transition: time 1 s
14. Y6 (Marking unit (stamping cylinder)) — close · transition: time 1 s
15. Y2 (Left clamping cylinder) — close · transition: feedback (watchdog time 4 s)
16. Y3 (Right clamping cylinder) — close · transition: feedback (watchdog time 4 s)
17. Y5 (Manipulator vacuum suction cup) — close · transition: time 1 s
18. M2 (Outfeed belt conveyor) — start · transition: feedback (watchdog time 3 s)

7. Fault behavior

Each standard device monitors its feedback signals with a timeout and reports a status: `16#0000` OK, `16#8001` inhibited (enable), `16#8002` fault. A block fault stops the device and is latched until acknowledged.

A **machine fault** (`machineFault`) is raised by a fault of any block or by expiry of the watchdog time of a sequence step (`faultStep` = step). The sequence immediately returns to step 0 and switches off all its commands; this also stops the other devices. A new start is possible only after the cause is removed and the fault acknowledged (`cmdAck`). The strategy (immediate stop × completing the step) is to be assessed according to the process.

8. Safety

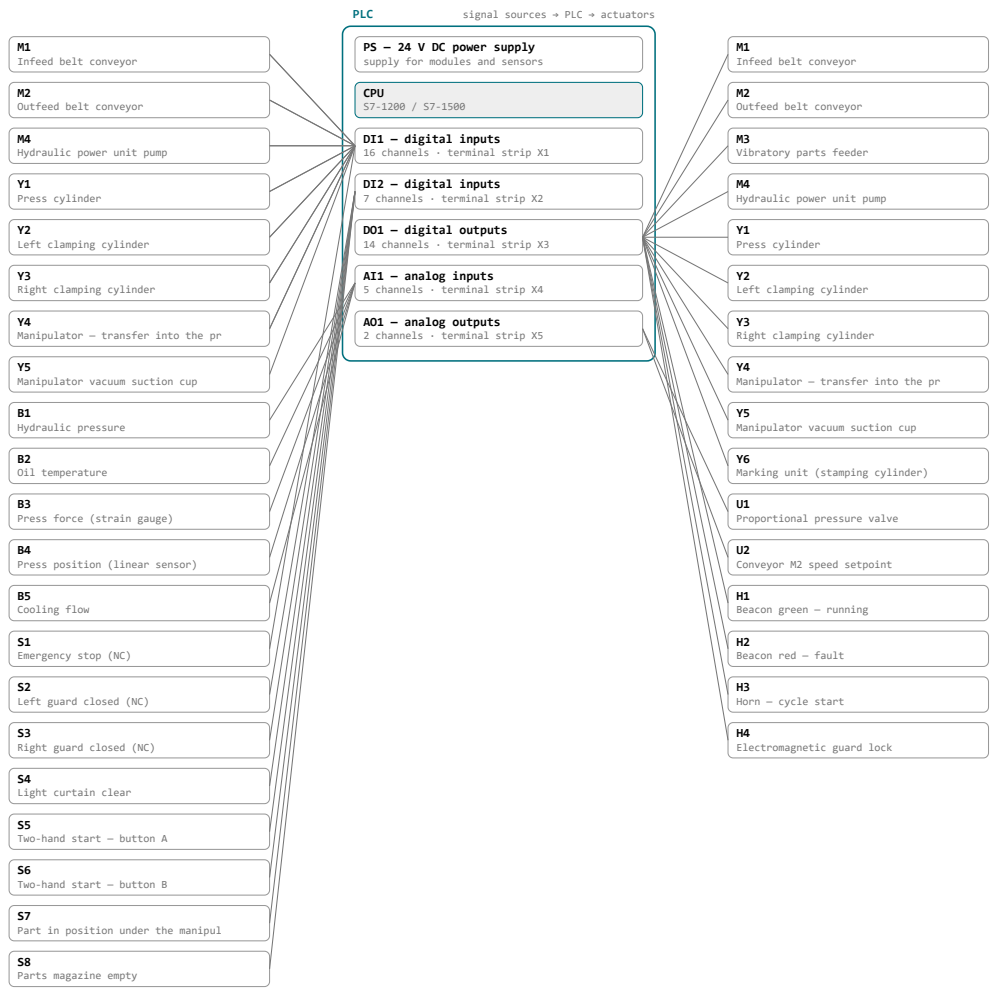
Safety functions (emergency stop, guards, two-hand control) are implemented by certified safety equipment according to the risk assessment (ISO 13849 / IEC 62061) — they are not part of this program. The signals in the PLC serve only to interlock the process and for diagnostics.

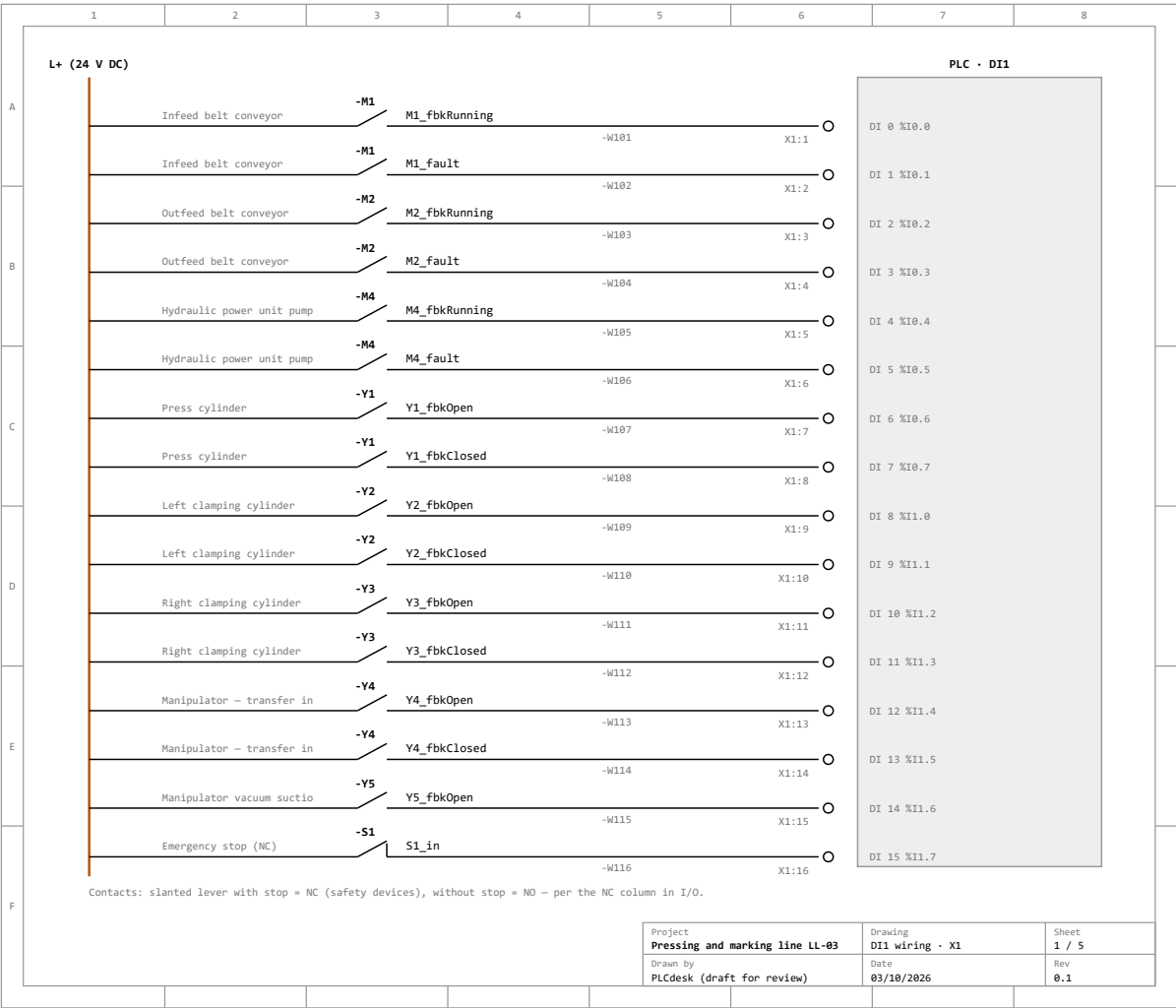
9. Out of scope of this document

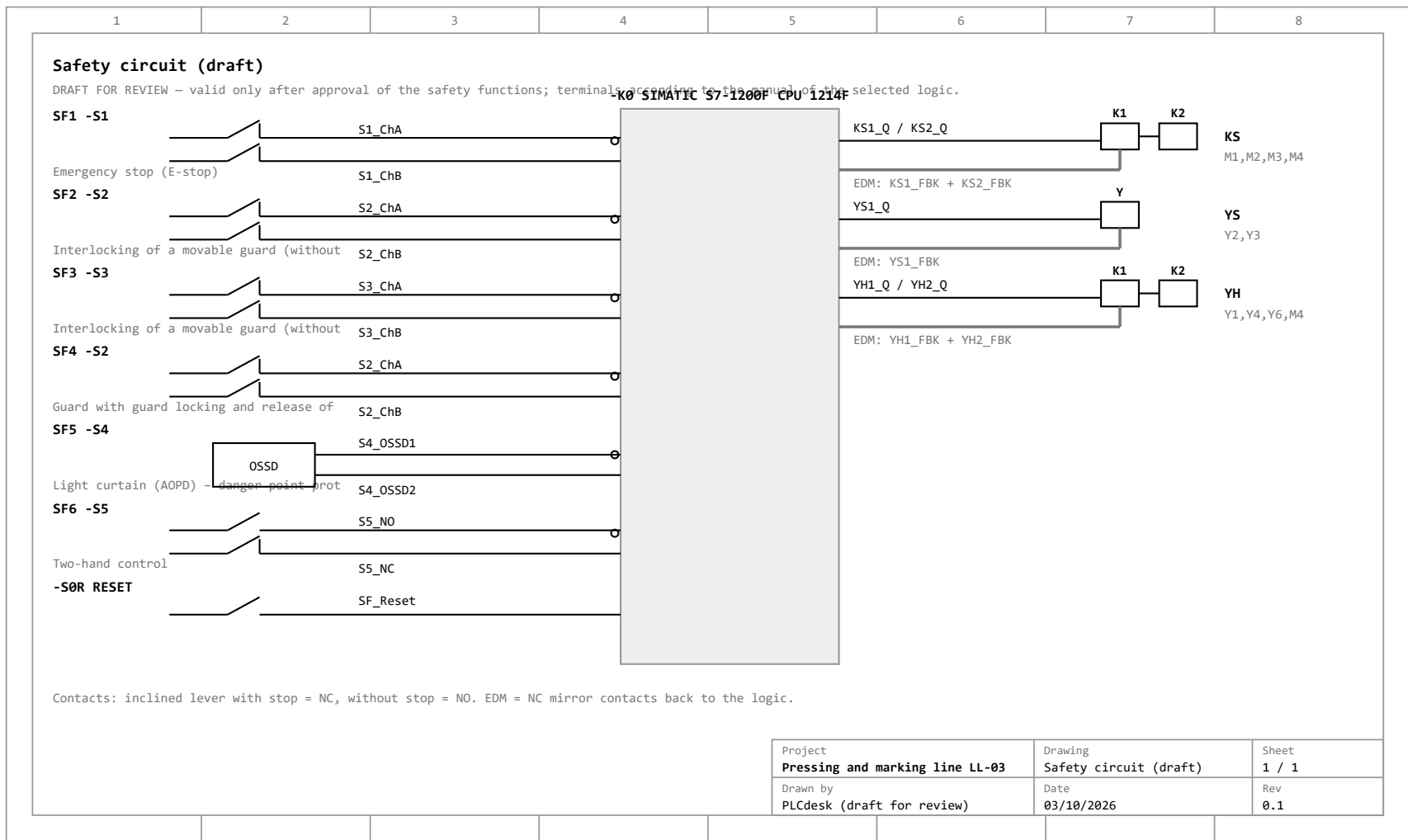
URS/FRS, risk assessment, HMI/SCADA specification, communication with higher-level systems, recipes.

Tag	Address	Direction	Data type	Device	Comment
M1_fbKRunning	%I0.0	DI	BOOL	M1	Infeed belt conveyor – running
M1_fault	%I0.1	DI	BOOL	M1	Infeed belt conveyor – fault
M1_outRun	%Q0.0	DO	BOOL	M1	Infeed belt conveyor – run command
M2_fbKRunning	%I0.2	DI	BOOL	M2	Outfeed belt conveyor – running
M2_fault	%I0.3	DI	BOOL	M2	Outfeed belt conveyor – fault
M2_outRun	%Q0.1	DO	BOOL	M2	Outfeed belt conveyor – run command
M3_outRun	%Q0.2	DO	BOOL	M3	Vibratory parts feeder – run command
M4_fbKRunning	%I0.4	DI	BOOL	M4	Hydraulic power unit pump – running
M4_fault	%I0.5	DI	BOOL	M4	Hydraulic power unit pump – fault
M4_outRun	%Q0.3	DO	BOOL	M4	Hydraulic power unit pump – run command
Y1_fbKOpen	%I0.6	DI	BOOL	Y1	Press cylinder – open
Y1_fbKClosed	%I0.7	DI	BOOL	Y1	Press cylinder – closed
Y1_outOpen	%Q0.4	DO	BOOL	Y1	Press cylinder – open command
Y2_fbKOpen	%I1.0	DI	BOOL	Y2	Left clamping cylinder – open
Y2_fbKClosed	%I1.1	DI	BOOL	Y2	Left clamping cylinder – closed
Y2_outOpen	%Q0.5	DO	BOOL	Y2	Left clamping cylinder – open command
Y3_fbKOpen	%I1.2	DI	BOOL	Y3	Right clamping cylinder – open
Y3_fbKClosed	%I1.3	DI	BOOL	Y3	Right clamping cylinder – closed
Y3_outOpen	%Q0.6	DO	BOOL	Y3	Right clamping cylinder – open command
Y4_fbKOpen	%I1.4	DI	BOOL	Y4	Manipulator — transfer into the press – open
Y4_fbKClosed	%I1.5	DI	BOOL	Y4	Manipulator — transfer into the press – closed
Y4_outOpen	%Q0.7	DO	BOOL	Y4	Manipulator — transfer into the press – open command
Y5_fbKOpen	%I1.6	DI	BOOL	Y5	Manipulator vacuum suction cup – open
Y5_outOpen	%Q1.0	DO	BOOL	Y5	Manipulator vacuum suction cup – open command
Y6_outOpen	%Q1.1	DO	BOOL	Y6	Marking unit (stamping cylinder) – open command
B1_raw	%IW64	AI	INT	B1	Hydraulic pressure – bar
B2_raw	%IW66	AI	INT	B2	Oil temperature – °C
B3_raw	%IW68	AI	INT	B3	Press force (strain gauge) – kN
B4_raw	%IW70	AI	INT	B4	Press position (linear sensor) – mm
B5_raw	%IW72	AI	INT	B5	Cooling flow – l/min
U1_raw	%QW64	AO	INT	U1	Proportional pressure valve – %
U2_raw	%QW66	AO	INT	U2	Conveyor M2 speed setpoint – %
S1_in	%I1.7	DI	BOOL	S1	Emergency stop (NC)
S2_in	%I2.0	DI	BOOL	S2	Left guard closed (NC)
S3_in	%I2.1	DI	BOOL	S3	Right guard closed (NC)
S4_in	%I2.2	DI	BOOL	S4	Light curtain clear
S5_in	%I2.3	DI	BOOL	S5	Two-hand start — button A
S6_in	%I2.4	DI	BOOL	S6	Two-hand start — button B
S7_in	%I2.5	DI	BOOL	S7	Part in position under the manipulator
S8_in	%I2.6	DI	BOOL	S8	Parts magazine empty
H1_out	%Q1.2	DO	BOOL	H1	Beacon green — running
H2_out	%Q1.3	DO	BOOL	H2	Beacon red — fault
H3_out	%Q1.4	DO	BOOL	H3	Horn — cycle start
H4_out	%Q1.5	DO	BOOL	H4	Electromagnetic guard lock

Terminal	Module	Channel	Address	Tag	Comment
X1:1	DI1	0	%I0.0	M1_fbkRunning	Infeed belt conveyor – running
X1:2	DI1	1	%I0.1	M1_fault	Infeed belt conveyor – fault
X1:3	DI1	2	%I0.2	M2_fbkRunning	Outfeed belt conveyor – running
X1:4	DI1	3	%I0.3	M2_fault	Outfeed belt conveyor – fault
X1:5	DI1	4	%I0.4	M4_fbkRunning	Hydraulic power unit pump – running
X1:6	DI1	5	%I0.5	M4_fault	Hydraulic power unit pump – fault
X1:7	DI1	6	%I0.6	Y1_fbkOpen	Press cylinder – open
X1:8	DI1	7	%I0.7	Y1_fbkClosed	Press cylinder – closed
X1:9	DI1	8	%I1.0	Y2_fbkOpen	Left clamping cylinder – open
X1:10	DI1	9	%I1.1	Y2_fbkClosed	Left clamping cylinder – closed
X1:11	DI1	10	%I1.2	Y3_fbkOpen	Right clamping cylinder – open
X1:12	DI1	11	%I1.3	Y3_fbkClosed	Right clamping cylinder – closed
X1:13	DI1	12	%I1.4	Y4_fbkOpen	Manipulator — transfer into the press – open
X1:14	DI1	13	%I1.5	Y4_fbkClosed	Manipulator — transfer into the press – closed
X1:15	DI1	14	%I1.6	Y5_fbkOpen	Manipulator vacuum suction cup – open
X1:16	DI1	15	%I1.7	S1_in	Emergency stop (NC)
X2:1	DI2	0	%I2.0	S2_in	Left guard closed (NC)
X2:2	DI2	1	%I2.1	S3_in	Right guard closed (NC)
X2:3	DI2	2	%I2.2	S4_in	Light curtain clear
X2:4	DI2	3	%I2.3	S5_in	Two-hand start — button A
X2:5	DI2	4	%I2.4	S6_in	Two-hand start — button B
X2:6	DI2	5	%I2.5	S7_in	Part in position under the manipulator
X2:7	DI2	6	%I2.6	S8_in	Parts magazine empty
X3:1	DO1	0	%Q0.0	M1_outRun	Infeed belt conveyor – run command
X3:2	DO1	1	%Q0.1	M2_outRun	Outfeed belt conveyor – run command
X3:3	DO1	2	%Q0.2	M3_outRun	Vibratory parts feeder – run command
X3:4	DO1	3	%Q0.3	M4_outRun	Hydraulic power unit pump – run command
X3:5	DO1	4	%Q0.4	Y1_outOpen	Press cylinder – open command
X3:6	DO1	5	%Q0.5	Y2_outOpen	Left clamping cylinder – open command
X3:7	DO1	6	%Q0.6	Y3_outOpen	Right clamping cylinder – open command
X3:8	DO1	7	%Q0.7	Y4_outOpen	Manipulator — transfer into the press – open command
X3:9	DO1	8	%Q1.0	Y5_outOpen	Manipulator vacuum suction cup – open command
X3:10	DO1	9	%Q1.1	Y6_outOpen	Marking unit (stamping cylinder) – open command
X3:11	DO1	10	%Q1.2	H1_out	Beacon green — running
X3:12	DO1	11	%Q1.3	H2_out	Beacon red — fault
X3:13	DO1	12	%Q1.4	H3_out	Horn — cycle start
X3:14	DO1	13	%Q1.5	H4_out	Electromagnetic guard lock
X4:1	AI1	0	%IW64	B1_raw	Hydraulic pressure – bar
X4:2	AI1	1	%IW66	B2_raw	Oil temperature – °C
X4:3	AI1	2	%IW68	B3_raw	Press force (strain gauge) – kN
X4:4	AI1	3	%IW70	B4_raw	Press position (linear sensor) – mm
X4:5	AI1	4	%IW72	B5_raw	Cooling flow – l/min
X5:1	AO1	0	%QW64	U1_raw	Proportional pressure valve – %
X5:2	AO1	1	%QW66	U2_raw	Conveyor M2 speed setpoint – %







Alarms

04_seznam_alarmu.csv

Code	Device	Alarm	Cause	System reaction	Acknowledge
A_M1_START	M1	Start-up timeout	Running feedback not received within 3 s	Device stop, machine fault (sequence stop)	Acknowledge (cmdAck) after the cause is removed
A_M1_FAULT	M1	External fault	Circuit breaker/drive reports a fault	Device stop, machine fault (sequence stop)	Acknowledge (cmdAck) after the fault has cleared
A_M1_RUN	M1	Loss of running feedback	Dropout during operation	Device stop, machine fault (sequence stop)	Acknowledge (cmdAck)
A_M2_START	M2	Start-up timeout	Running feedback not received within 3 s	Device stop, machine fault (sequence stop)	Acknowledge (cmdAck) after the cause is removed
A_M2_FAULT	M2	External fault	Circuit breaker/drive reports a fault	Device stop, machine fault (sequence stop)	Acknowledge (cmdAck) after the fault has cleared
A_M2_RUN	M2	Loss of running feedback	Dropout during operation	Device stop, machine fault (sequence stop)	Acknowledge (cmdAck)
A_M3_START	M3	Start-up timeout	Running feedback not received within 3 s	Device stop, machine fault (sequence stop)	Acknowledge (cmdAck) after the cause is removed
A_M4_START	M4	Start-up timeout	Running feedback not received within 3 s	Device stop, machine fault (sequence stop)	Acknowledge (cmdAck) after the cause is removed
A_M4_FAULT	M4	External fault	Circuit breaker/drive reports a fault	Device stop, machine fault (sequence stop)	Acknowledge (cmdAck) after the fault has cleared
A_M4_RUN	M4	Loss of running feedback	Dropout during operation	Device stop, machine fault (sequence stop)	Acknowledge (cmdAck)
A_Y1_TRAVEL	Y1	Travel timeout	End position not reached within 5 s	Output switched off, machine fault (sequence stop)	Acknowledge (cmdAck)
A_Y1_POS	Y1	Loss of open position	The "open" limit switch dropped out in the held position	Output switched off, machine fault (sequence stop)	Acknowledge (cmdAck) after the cause is removed
A_Y2_TRAVEL	Y2	Travel timeout	End position not reached within 5 s	Output switched off, machine fault (sequence stop)	Acknowledge (cmdAck)
A_Y2_POS	Y2	Loss of open position	The "open" limit switch dropped out in the held position	Output switched off, machine fault (sequence stop)	Acknowledge (cmdAck) after the cause is removed
A_Y3_TRAVEL	Y3	Travel timeout	End position not reached within 5 s	Output switched off, machine fault (sequence stop)	Acknowledge (cmdAck)
A_Y3_POS	Y3	Loss of open position	The "open" limit switch dropped out in the held position	Output switched off, machine fault (sequence stop)	Acknowledge (cmdAck) after the cause is removed
A_Y4_TRAVEL	Y4	Travel timeout	End position not reached within 5 s	Output switched off, machine fault (sequence stop)	Acknowledge (cmdAck)
A_Y4_POS	Y4	Loss of open position	The "open" limit switch dropped out in the held position	Output switched off, machine fault (sequence stop)	Acknowledge (cmdAck) after the cause is removed
A_Y5_TRAVEL	Y5	Travel timeout	End position not reached within 5 s	Output switched off, machine fault (sequence stop)	Acknowledge (cmdAck)
A_Y5_POS	Y5	Loss of open position	The "open" limit switch dropped out in the held position	Output switched off, machine fault (sequence stop)	Acknowledge (cmdAck) after the cause is removed
A_Y6_TRAVEL	Y6	Travel timeout	End position not reached within 5 s	Output switched off, machine fault (sequence stop)	Acknowledge (cmdAck)
A_B1_HI	B1	High limit exceeded	Value > limitHi (bar)	Depending on the process (to be completed)	Automatic on return to normal
A_B1_LO	B1	Below low limit	Value < limitLo (bar)	Depending on the process (to be completed)	Automatic on return to normal
A_B2_HI	B2	High limit exceeded	Value > limitHi (°C)	Depending on the process (to be completed)	Automatic on return to normal
A_B2_LO	B2	Below low limit	Value < limitLo (°C)	Depending on the process (to be completed)	Automatic on return to normal
A_B3_HI	B3	High limit exceeded	Value > limitHi (kN)	Depending on the process (to be completed)	Automatic on return to normal
A_B3_LO	B3	Below low limit	Value < limitLo (kN)	Depending on the process (to be completed)	Automatic on return to normal
A_B4_HI	B4	High limit exceeded	Value > limitHi (mm)	Depending on the process (to be completed)	Automatic on return to normal
A_B4_LO	B4	Below low limit	Value < limitLo (mm)	Depending on the process (to be completed)	Automatic on return to normal
A_B5_HI	B5	High limit exceeded	Value > limitHi (l/min)	Depending on the process (to be completed)	Automatic on return to normal
A_B5_LO	B5	Below low limit	Value < limitLo (l/min)	Depending on the process (to be completed)	Automatic on return to normal
A_S2_OPEN	S2	Interlock open	Left guard closed (NC): input FALSE (guard open / light curtain interrupted)	Machine stop: block outputs off, sequence to step 0	New start after restoring (cmdAutoStart)

Code	Device	Alarm	Cause	System reaction	Acknowledge
A_S3_OPEN	S3	Interlock open	Right guard closed (NC): input FALSE (guard open / light curtain interrupted)	Machine stop: block outputs off, sequence to step 0	New start after restoring (cmdAutoStart)
A_S4_OPEN	S4	Interlock open	Light curtain clear: input FALSE (guard open / light curtain interrupted)	Machine stop: block outputs off, sequence to step 0	New start after restoring (cmdAutoStart)
A_SEQ_2	sequence	Step 2 timeout (M1 start)	Condition "M1_fbkRunning" not met within 3 s	Machine fault: sequence to step 0, commands off (faultStep = 20)	Acknowledge (cmdAck) after the cause is removed
A_SEQ_3	sequence	Step 3 timeout (Y4 open)	Condition "Y4_fbkOpen" not met within 5 s	Machine fault: sequence to step 0, commands off (faultStep = 30)	Acknowledge (cmdAck) after the cause is removed
A_SEQ_4	sequence	Step 4 timeout (Y5 open)	Condition "Y5_fbkOpen" not met within 2 s	Machine fault: sequence to step 0, commands off (faultStep = 40)	Acknowledge (cmdAck) after the cause is removed
A_SEQ_5	sequence	Step 5 timeout (Y4 close)	Condition "Y4_fbkClosed" not met within 5 s	Machine fault: sequence to step 0, commands off (faultStep = 50)	Acknowledge (cmdAck) after the cause is removed
A_SEQ_6	sequence	Step 6 timeout (Y2 open)	Condition "Y2_fbkOpen" not met within 4 s	Machine fault: sequence to step 0, commands off (faultStep = 60)	Acknowledge (cmdAck) after the cause is removed
A_SEQ_7	sequence	Step 7 timeout (Y3 open)	Condition "Y3_fbkOpen" not met within 4 s	Machine fault: sequence to step 0, commands off (faultStep = 70)	Acknowledge (cmdAck) after the cause is removed
A_SEQ_8	sequence	Step 8 timeout (M4 start)	Condition "M4_fbkRunning" not met within 3 s	Machine fault: sequence to step 0, commands off (faultStep = 80)	Acknowledge (cmdAck) after the cause is removed
A_SEQ_9	sequence	Step 9 timeout (Y1 open)	Condition "Y1_fbkOpen" not met within 8 s	Machine fault: sequence to step 0, commands off (faultStep = 90)	Acknowledge (cmdAck) after the cause is removed
A_SEQ_11	sequence	Step 11 timeout (Y1 close)	Condition "Y1_fbkClosed" not met within 8 s	Machine fault: sequence to step 0, commands off (faultStep = 110)	Acknowledge (cmdAck) after the cause is removed
A_SEQ_12	sequence	Step 12 timeout (M4 stop)	Condition "NOT M4_fbkRunning" not met within 3 s	Machine fault: sequence to step 0, commands off (faultStep = 120)	Acknowledge (cmdAck) after the cause is removed
A_SEQ_15	sequence	Step 15 timeout (Y2 close)	Condition "Y2_fbkClosed" not met within 4 s	Machine fault: sequence to step 0, commands off (faultStep = 150)	Acknowledge (cmdAck) after the cause is removed
A_SEQ_16	sequence	Step 16 timeout (Y3 close)	Condition "Y3_fbkClosed" not met within 4 s	Machine fault: sequence to step 0, commands off (faultStep = 160)	Acknowledge (cmdAck) after the cause is removed
A_SEQ_18	sequence	Step 18 timeout (M2 start)	Condition "M2_fbkRunning" not met within 3 s	Machine fault: sequence to step 0, commands off (faultStep = 180)	Acknowledge (cmdAck) after the cause is removed

FAT test protocol

CHANGED AFTER APPROVAL — content changed after approval: Automatic sequence, Cycle time target and watchdog times. The approval is no longer valid — approve again (see 11_schvaleni.md).

Project: Pressing and marking line LL-03 (complex example) · **Test date:** · **Tested by:** · **For the customer:**

Tick ☐ → ☒. Record nonconformities in the table at the end. SAT = the same scope after installation at the customer's site + the real actuators and process.

1. I/O loop check

Terminal	Address	Tag	Procedure	OK
X1:1	%I0.0	M1_fbkRunning	Actuate the sensor/contact, verify in the PLC	☐
X1:2	%I0.1	M1_fault	Actuate the sensor/contact, verify in the PLC	☐
X1:3	%I0.2	M2_fbkRunning	Actuate the sensor/contact, verify in the PLC	☐
X1:4	%I0.3	M2_fault	Actuate the sensor/contact, verify in the PLC	☐
X1:5	%I0.4	M4_fbkRunning	Actuate the sensor/contact, verify in the PLC	☐
X1:6	%I0.5	M4_fault	Actuate the sensor/contact, verify in the PLC	☐
X1:7	%I0.6	Y1_fbkOpen	Actuate the sensor/contact, verify in the PLC	☐
X1:8	%I0.7	Y1_fbkClosed	Actuate the sensor/contact, verify in the PLC	☐
X1:9	%I1.0	Y2_fbkOpen	Actuate the sensor/contact, verify in the PLC	☐
X1:10	%I1.1	Y2_fbkClosed	Actuate the sensor/contact, verify in the PLC	☐
X1:11	%I1.2	Y3_fbkOpen	Actuate the sensor/contact, verify in the PLC	☐
X1:12	%I1.3	Y3_fbkClosed	Actuate the sensor/contact, verify in the PLC	☐
X1:13	%I1.4	Y4_fbkOpen	Actuate the sensor/contact, verify in the PLC	☐
X1:14	%I1.5	Y4_fbkClosed	Actuate the sensor/contact, verify in the PLC	☐
X1:15	%I1.6	Y5_fbkOpen	Actuate the sensor/contact, verify in the PLC	☐
X1:16	%I1.7	S1_in	Actuate the sensor/contact, verify in the PLC	☐
X2:1	%I2.0	S2_in	Actuate the sensor/contact, verify in the PLC	☐
X2:2	%I2.1	S3_in	Actuate the sensor/contact, verify in the PLC	☐
X2:3	%I2.2	S4_in	Actuate the sensor/contact, verify in the PLC	☐
X2:4	%I2.3	S5_in	Actuate the sensor/contact, verify in the PLC	☐
X2:5	%I2.4	S6_in	Actuate the sensor/contact, verify in the PLC	☐
X2:6	%I2.5	S7_in	Actuate the sensor/contact, verify in the PLC	☐
X2:7	%I2.6	S8_in	Actuate the sensor/contact, verify in the PLC	☐
X3:1	%Q0.0	M1_outRun	Force the output from the PLC, verify the actuator	☐
X3:2	%Q0.1	M2_outRun	Force the output from the PLC, verify the actuator	☐
X3:3	%Q0.2	M3_outRun	Force the output from the PLC, verify the actuator	☐
X3:4	%Q0.3	M4_outRun	Force the output from the PLC, verify the actuator	☐
X3:5	%Q0.4	Y1_outOpen	Force the output from the PLC, verify the actuator	☐
X3:6	%Q0.5	Y2_outOpen	Force the output from the PLC, verify the actuator	☐
X3:7	%Q0.6	Y3_outOpen	Force the output from the PLC, verify the actuator	☐
X3:8	%Q0.7	Y4_outOpen	Force the output from the PLC, verify the actuator	☐
X3:9	%Q1.0	Y5_outOpen	Force the output from the PLC, verify the actuator	☐
X3:10	%Q1.1	Y6_outOpen	Force the output from the PLC, verify the actuator	☐
X3:11	%Q1.2	H1_out	Force the output from the PLC, verify the actuator	☐
X3:12	%Q1.3	H2_out	Force the output from the PLC, verify the actuator	☐
X3:13	%Q1.4	H3_out	Force the output from the PLC, verify the actuator	☐
X3:14	%Q1.5	H4_out	Force the output from the PLC, verify the actuator	☐

Terminal	Address	Tag	Procedure	OK
X4:1	%IW64	B1_raw	Signal source (calibrator), verify the value and scaling	☐
X4:2	%IW66	B2_raw	Signal source (calibrator), verify the value and scaling	☐
X4:3	%IW68	B3_raw	Signal source (calibrator), verify the value and scaling	☐
X4:4	%IW70	B4_raw	Signal source (calibrator), verify the value and scaling	☐
X4:5	%IW72	B5_raw	Signal source (calibrator), verify the value and scaling	☐
X5:1	%QW64	U1_raw	Force a value, measure the output	☐
X5:2	%QW66	U2_raw	Force a value, measure the output	☐

2. Device functional tests

M1 — Infeed belt conveyor

- Start → runs, running feedback within 3 s ☐
- Stop → stops ☐
- Simulated loss of feedback during operation → fault, output off ☐
- Simulated external fault → fault ☐
- Fault acknowledge ☐

M2 — Outfeed belt conveyor

- Start → runs, running feedback within 3 s ☐
- Stop → stops ☐
- Simulated loss of feedback during operation → fault, output off ☐
- Simulated external fault → fault ☐
- Fault acknowledge ☐

M3 — Vibratory parts feeder

- Start → runs, running feedback within 3 s ☐
- Stop → stops ☐
- Simulated loss of feedback during operation → fault, output off ☐
- Fault acknowledge ☐

M4 — Hydraulic power unit pump

- Start → runs, running feedback within 3 s ☐
- Stop → stops ☐
- Simulated loss of feedback during operation → fault, output off ☐
- Simulated external fault → fault ☐
- Fault acknowledge ☐

Y1 — Press cylinder

- Open → open position reached ☐
- Close → closed position reached ☐
- Simulated failure to reach position → timeout, fault ☐
- Acknowledge ☐

Y2 — Left clamping cylinder

- Open → open position reached ☐
- Close → closed position reached ☐
- Simulated failure to reach position → timeout, fault ☐
- Acknowledge ☐

Y3 — Right clamping cylinder

- Open → open position reached ☐
- Close → closed position reached ☐
- Simulated failure to reach position → timeout, fault ☐
- Acknowledge ☐

Y4 — Manipulator — transfer into the press

- Open → open position reached ☐
- Close → closed position reached ☐
- Simulated failure to reach position → timeout, fault ☐
- Acknowledge ☐

Y5 — Manipulator vacuum suction cup

- Open → open position reached ☐
- Close → closed position reached ☐
- Simulated failure to reach position → timeout, fault ☐
- Acknowledge ☐

Y6 — Marking unit (stamping cylinder)

- Open → open position reached ☐
- Close → closed position reached ☐
- Simulated failure to reach position → timeout, fault ☐
- Acknowledge ☐

B1 — Hydraulic pressure (bar 0–250)

- Comparison with the actual/calibrated value at 3 points ☐
- Hi/Lo limit test ☐

B2 — Oil temperature (°C 0–100)

- Comparison with the actual/calibrated value at 3 points ☐
- Hi/Lo limit test ☐

B3 — Press force (strain gauge) (kN 0–400)

- Comparison with the actual/calibrated value at 3 points ☐
- Hi/Lo limit test ☐

B4 — Press position (linear sensor) (mm 0–500)

- Comparison with the actual/calibrated value at 3 points ☐
- Hi/Lo limit test ☐

B5 — Cooling flow (l/min 0–60)

- Comparison with the actual/calibrated value at 3 points ☐
- Hi/Lo limit test ☐

3. Automatic sequence test

Step	Expected behavior	OK
1	M3 start → transition on time	☐
2	M1 start → transition on feedback	☐
3	Y4 open → transition on feedback	☐
4	Y5 open → transition on feedback	☐
5	Y4 close → transition on feedback	☐
6	Y2 open → transition on feedback	☐
7	Y3 open → transition on feedback	☐
8	M4 start → transition on feedback	☐
9	Y1 open → transition on feedback	☐
10	dwelt 5 s → transition on time	☐
11	Y1 close → transition on feedback	☐
12	M4 stop → transition on feedback	☐
13	Y6 open → transition on time	☐
14	Y6 close → transition on time	☐
15	Y2 close → transition on feedback	☐
16	Y3 close → transition on feedback	☐
17	Y5 close → transition on time	☐
18	M2 start → transition on feedback	☐
—	Central enable interrupted mid-cycle → everything stopped safely	☐
—	Simulated block fault / failure to reach position in a step → machine fault, sequence to step 0, commands off	☐
—	Start without acknowledge → cycle does not start	☐
—	Restart after acknowledge → cycle from the beginning	☐

Step	Expected behavior	OK
—	Manual mode: commands to individual devices from the HMI; no effect in AUTO	☐

4. Nonconformities

No.	Description	Severity	Resolved

FAT result: PASSED / PASSED WITH RESERVATIONS / FAILED

Signatures:

Operating manual — Pressing and marking line LL-03 (complex example)

CHANGED AFTER APPROVAL — content changed after approval: Automatic sequence, Cycle time target and watchdog times. The approval is no longer valid — approve again (see 11_schvaleni.md).

Revision 0.1 (03/10/2026) — skeleton to be completed; add photographs, the control panel and contacts before handover.

1. Machine description

Parts feeder, vacuum manipulator, hydraulic press with marking and outfeed conveyor; two-hand start, light curtain, two guards.

2. Controls

(to be completed: main switch, HMI panel, push buttons, indicators — H1 Beacon green — running, H2 Beacon red — fault, H3 Horn — cycle start, H4 Electromagnetic guard lock)

3. Start-up

1. Switch on the main switch, check the indicators.
2. Release the emergency stop (S1 – Emergency stop (NC)), close the guards.
3. Select MANUAL/AUTO mode.
4. In AUTO mode, start the cycle with the start button.

4. Stopping

- Normal: stop button / completion of the cycle.
- Emergency: emergency stop button — ONLY in case of danger; acknowledge required after use.

5. Faults and troubleshooting

Alarm list: see 04_seznam_alarmu.csv . General procedure: remove the cause → acknowledge → restart. If the fault recurs, contact maintenance.

6. Maintenance

(add intervals: lubrication, sensor checks, terminal tightening, analog calibration — recommended once a year)

7. Safety notices

Work on the electrical equipment may only be carried out by a suitably qualified person. Safety devices must not be disabled or bypassed.

Safety functions — specification (SRS)

NOT APPROVED — 13 of 22 items lack approval: SF5 — Light curtain (AOPD) – danger point protection (finger/hand) — S4, SF5 — architecture, components and PL calculation, SF6 — Two-hand control — S5, S6, SF6 — architecture, components and PL calculation, SF7 — Prevention of unexpected start-up (isolation and energy dissipation, LOTO, start interlock), SF7 — architecture, components and PL calculation, ... (+7). Draft for review — do not use as a basis without approval (see 11_schvaleni.md).

Project: Pressing and marking line LL-03 (complex example) · **Created:** 03/10/2026 · **Research data:** 2026-10-03

PLCdesk PROPOSES hazards, safety functions, PLr, architecture, components and the PL calculation; only what the responsible person approves is valid (name, date, note — 11_schvaleni.md). The proposal is based on ISO 13849-1 (risk graph annex A, simplified PL determination), IEC 60204-1 (stop categories) and ISO 13855 (distances); the standards are paid, the data come from public interpretations (IFA/DGUV, manufacturers) — sources are given for each function. Draft for review; it does not replace the standard or the risk assessment.

Safety logic belongs in a safety PLC / relay. The standard program reads E-stop and interlocking only as status signals (enable, acknowledgement) — safety functions do not rely on it.

Design parameters

Parameter	Value
Safety logic	Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC (6ES7214-1AF40-0XB0)
Safety program target	Siemens F-program
Operating days / hours (nop)	220 d · 16 h
Mission time	20 years
ISO 13855 edition	2010
PL method	simplified: category × MTTFd × DCavg × CCF (ISO 13849-1 bar chart); PL of the function = lowest PL of the subsystems — verify in SISTEMA (PFH by summation)

1. Hazards and assigned functions

Approval status: **approved**

Hazard	Device	Function
Emergency situation at any hazard of the machine	—	SF1
Mechanical hazard from moving parts of drives (drawing-in, entanglement, impact, crushing)	M1, M2, M3, M4	SF2, SF3, SF4, SF5
Crushing and shearing by pneumatic cylinders; residual energy of compressed air	Y2, Y3	SF2, SF3, SF4, SF5, SF8
Crushing by hydraulic drive, ejection of pressurised fluid, residual pressure	Y1, Y4, Y6, M4	SF2, SF3, SF4, SF5, SF6, SF9, SF10
Crushing and shearing in the press working area (ram / tool stroke)	Y1, Y4, Y6, B3, B4	SF2, SF3, SF4, SF5, SF6
Entering or reaching into the hazardous area during operation	S2, S3, S2, S3, H4, S4, S5, S6	SF2, SF3, SF4, SF5, SF6
Unexpected start-up after intervention, reset or restoration of energy	—	SF7
Overpressure in a pressure system (rupture, escape of medium)	—	SF10

2. Overview of safety functions

ID	Function	PLr	PL achieved	Cat.	Stop	Function approval	Design approval
SF1	Emergency stop (E-stop) — S1	d	d ✓	3	0	approved	approved
SF2	Interlocking of a movable guard (without guard locking) — S2	e	e ✓	4	0	approved	approved
SF3	Interlocking of a movable guard (without guard locking) — S3	e	e ✓	4	0	approved	approved
SF4	Guard with guard locking and release of the lock after stopping — S2, S3, H4	d	d ✓	3	0	approved	approved
SF5	Light curtain (AOPD) – danger point protection (finger/hand) — S4	e	— ✗	4	0	not approved	not approved
SF6	Two-hand control — S5, S6	e	e ✓	4	0	not approved	not approved
SF7	Prevention of unexpected start-up (isolation and energy dissipation, LOTO, start interlock)	d	d ✓	3	0	not approved	not approved
SF8	Safe pneumatic exhaust (SDE) and blocking of pneumatic movements (SSC)	e	e ✓	4	0	not approved	not approved
SF9	Safe stop / isolation of hydraulics (redundant valves with position monitoring)	e	e ✓	4	0	not approved	not approved

ID	Function	PLr	PL achieved	Cat.	Stop	Function approval	Design approval
SF10	Pressure limitation (relief valve) and pressure monitoring	—	— ×	B	0	not approved	not approved

Safety program: **not approved**

3. Function specifications

SF1 — Emergency stop (E-stop) — S1

Status: function **approved**, design **approved**

SRS field	Value
Description (input → logic → output)	S1 → SIMATIC S7-1200F CPU 1214FC DC/DC/DC → KS, YS, YH
Hazard	Emergency situation at any hazard of the machine
Triggering event	Pressing the mushroom button / pulling the emergency stop rope (direct opening action, mechanical latching). Inputs: S1 (Emergency stop (NC)).
Reaction	Terminate the hazardous state as quickly as possible without creating further risks; the command takes precedence over all other functions in all modes; the state is maintained until the actuator is manually released and reset; releasing must not start the machine. Outputs: contactors KS1/KS2 with mirror contacts for M1, M2, M3, M4; safety exhaust valve YS1 for Y2, Y3; redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4.
Safe state	drives de-energised (contactors dropped out / STO); pneumatics exhausted; hydraulics depressurised, movement blocked
Stop category (IEC 60204-1)	0 — stop category 0: immediate removal of energy (contactors, STO, exhaust)
PLr	d — risk graph S2 / F1 / P2 → d; normative minimum c: ISO 13850:2015 requires at least PL c for emergency stop.
Operating modes	all operating modes (AUTO and MANUAL), priority over all commands
Permissible response time	to be completed — at most the time within which a person cannot reach the hazard (ISO 13855)
Demand rate	on average once every 8 h (assumption for nop — verify according to operation)
Priority	highest — takes precedence over all commands and modes
Reset and restart	Release the actuator manually on site; reset only prepares the machine for restarting, start by a separate command. For a rope, the entire length of the rope must be visible from the reset location.
Interface to the standard program	S1_in
Fault reaction	Detected fault (channel discrepancy, short circuit, EDM) → safe state; reset is not possible while the fault persists.
Behaviour on loss of power	Loss of power = safe state (de-energise to trip, outputs de-energised); after power is restored a manual reset is required (manual cold start).
Fault exclusions	none proposed — justify every exclusion (ISO 13849-2); PL e must not rely on fault exclusions alone
Standards	ISO 13850, IEC 60204-1, ISO 13849-1, ISO 13849-2, IEC 60947-5-5, IEC 60947-5-1, ISO 12100, IEC 62061

Justification of S / F / P (proposal)

- S2: movements of the machine drives can cause serious, normally irreversible injury (fractures, amputation).
- F1: demand is rare and exposure time short (less than once per 15 min, cumulatively below 1/20 of the operating time).
- P2: in an emergency the operator usually cannot avoid the hazard.
- Typical according to sources: PLr c (minimum according to ISO 13850:2015), often d; architecture 1 (PL c) or 3 (PL d/e)

Architecture and PL calculation

Category 3, 2 channel(s), EDM yes, manual reset; CCF 70 points (met ≥ 65).

Subsystem	Component	Cat.	nop [1/year]	MTTFd [years]	DC [%]	PL	Notes
I: emergency stop button, 2 NC contacts with direct opening action	Siemens SIRIUS ACT 3SU1 emergency stop (3SU1150-1HB20-1CH0)	3	440	11363.6	99	d	MTTFd = B10d / (0.1 · nop) = 500000 / (0.1 · 440) = 11363.6 years; value from typical values / research — confirm against the datasheet of the specific variant; Channel MTTFd capped at 100 years
L: safety logic: Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC	Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC (6ES7214-1AF40-0XB0)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —
O: contactors KS1/KS2 with mirror contacts for M1, M2, M3, M4 — 2 contactors in series, EDM	Siemens SIRIUS 3RT20 / 3RT2	3	440	31136.4	99	d	MTTFd = B10d / (0.1 · nop) = 1370000 / (0.1 · 440) = 31136.4 years; value from typical values / research — confirm against the datasheet of the specific variant; Channel MTTFd capped at 100 years

Subsystem	Component	Cat.	nop [1/year]	MTTFd [years]	DC [%]	PL	Notes
O: safety exhaust valve YS1 for Y2, Y3	Festo MS6-SV-E (562580)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —
O: redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4	Hydraulic components, nop < 250 000 cycles/year	3	440	1200	99	d	value from typical values / research — confirm against the datasheet of the specific variant; Channel MTTFd capped at 100 years

PL achieved: d — PLr d met

Measures against CCF (to be confirmed by the user): Physical separation of the signal paths of different channels (15); Protection against possible overload (15); Use of well-tried components (5); Protection against CCF due to contamination (mechanics, fluid power systems) and electromagnetic interference (electrical systems) (25); Protection against CCF due to unfavourable environmental conditions (10)

Wiring

- Inputs S1_ChA, S1_ChB: dual-channel into the safety logic, pulsed signals (pulse test) for cross-circuit detection, channel discrepancy monitoring.
- Reset SF_Reset: button outside the hazardous area with a view into it; falling edge evaluation; reset only prepares, the start is given by the standard program.
- KS: two contactors in series controlled by outputs KS1_Q / KS2_Q; NC mirror contacts into EDM KS1_FBK / KS2_FBK (feedback before every start).
- YS: safety valve (supply exhaust / cylinder double valve), coil from output YS1_Q, position feedback YS1_FBK.
- YH: two valves in series, coils YH1_Q / YH2_Q, spool position monitoring YH1_FBK / YH2_FBK.
- Status to the standard PLC: S1_in (auxiliary contact / mirroring from the safety logic) — the standard program reads it only as enable and acknowledgement.

Signal	Direction	Meaning
S1_ChA	input	S1: channel A (NC contact)
S1_ChB	input	S1: channel B (NC contact)
SF_Reset	input	safety functions reset button (outside the hazardous area, with a view into it; falling edge)
KS1_Q	output	KS: safe output 1
KS2_Q	output	KS: safe output 2
KS1_FBK	input	KS: feedback (EDM / position)
KS2_FBK	input	KS: feedback (EDM / position)
YS1_Q	output	YS: safe output 1
YS1_FBK	input	YS: feedback (EDM / position)
YH1_Q	output	YH: safe output 1
YH2_Q	output	YH: safe output 2
YH1_FBK	input	YH: feedback (EDM / position)
YH2_FBK	input	YH: feedback (EDM / position)
S1_in	output	status of S1 to the standard PLC (information signal only, enable / acknowledgement)

Typical components according to sources

- emergency stop button according to ISO 13850 / IEC 60947-5-5 (red on yellow, latching, direct opening action according to IEC 60947-5-1 Annex K)
- rope pull switch (alternative)
- safety module or safety PLC with pulse testing of inputs
- 2 contactors with mechanically linked (force-guided) contacts or drive STO

Sources

- <https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf>
- <https://publikationen.dguv.de/widgets/pdf/download/article/5091>
- https://maxter-automatyka.pl/image/plik_o/960_bezpieczenstwo-maszyn-sick.pdf
- <https://www.idec.com/en-eu/solutions/safety/law/iso-iec/iso13850>
- <https://machinerysafety101.com/2026/05/18/iso-13850-emergency-stop-requirements/>
- https://literature.rockwellautomation.com/idc/groups/literature/documents/at/safety-at128_-en-p.pdf
- https://literature.rockwellautomation.com/idc/groups/literature/documents/at/safety-at203_-en-p.pdf
- https://literature.rockwellautomation.com/idc/groups/literature/documents/at/safety-at122_-en-p.pdf

SF2 — Interlocking of a movable guard (without guard locking) — S2

Status: function **approved**, design **approved**

SRS field	Value
Description (input → logic → output)	S2 → SIMATIC S7-1200F CPU 1214FC DC/DC/DC → KS, YS, YH
Hazard	Mechanical hazard from moving parts of drives (drawing-in, entanglement, impact, crushing); Crushing and shearing by pneumatic cylinders; residual energy of compressed air; Crushing by hydraulic drive, ejection of pressurised fluid, residual pressure; Crushing and shearing in the press working area (ram / tool stroke); Entering or reaching into the hazardous area during operation
Triggering event	Opening of the guard (leaving the 'closed' position detected by the switch). Inputs: S2 (Left guard closed (NC)).
Reaction	Safety stop (STO, SS1 or SS2) of hazardous movements; start blocked while the guard is open; after closing only manual reset/start (automatic restart only for control guards according to ISO 12100). Outputs: contactors KS1/KS2 with mirror contacts for M1, M2, M3, M4; safety exhaust valve YS1 for Y2, Y3; redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4.
Safe state	drives de-energised (contactors dropped out / STO); pneumatics exhausted; hydraulics depressurised, movement blocked
Stop category (IEC 60204-1)	0 — stop category 0: immediate removal of energy (contactors, STO, exhaust)
PLr	e — risk graph S2 / F2 / P2 → e
Operating modes	all operating modes (AUTO and MANUAL) — the standard program does not bypass the protection
Permissible response time	to be completed — at most the time within which a person cannot reach the hazard (ISO 13855)
Demand rate	on average once every 15 min (assumption for nop — verify according to operation)
Priority	stop takes precedence over start; emergency stop over all functions
Reset and restart	Manual reset with button SF_Reset outside the hazardous area (falling edge, monitored signal change); reset does not start the machine — a new start only by the START command of the standard program.
Interface to the standard program	S2_in
Fault reaction	Detected fault (channel discrepancy, short circuit, EDM) → safe state; reset is not possible while the fault persists.
Behaviour on loss of power	Loss of power = safe state (de-energise to trip, outputs de-energised); after power is restored a manual reset is required (manual cold start).
Fault exclusions	none proposed — justify every exclusion (ISO 13849-2); PL e must not rely on fault exclusions alone
Standards	ISO 14119, ISO 13849-1, ISO 13849-2, ISO 13855, ISO 13857, ISO 14120, IEC 60947-5-1, IEC 60947-5-3, IEC 60204-1

Justification of S / F / P (proposal)

- S2: crushing / shearing in the press tool — serious irreversible injury.
- F2: operator interventions (loading, troubleshooting, cleaning) may be more frequent than once per 15 min — adjust according to actual operation.
- P2: the press stroke is fast, a hand in the tool cannot be withdrawn in time.
- Typical according to sources: PLr d; architecture 3 (PL d) typically

Architecture and PL calculation

Category 4, 2 channel(s), EDM yes, manual reset; CCF 70 points (met ≥ 65).

Subsystem	Component	Cat.	nop [1/year]	MTTFd [years]	DC [%]	PL	Notes
I: coded non-contact guard switch (RFID, OSSD)	Schmersal RSS 260 (103003602)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd 6,8E-10 1/h
L: safety logic: Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC	Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC (6ES7214-1AF40-0XB0)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —
O: contactors KS1/KS2 with mirror contacts for M1, M2, M3, M4 — 2 contactors in series, EDM	Siemens SIRIUS 3RT20 / 3RT2	4	14080	973	99	e	MTTFd = B10d / (0.1 · nop) = 1370000 / (0.1 · 14080) = 973 years; value from typical values / research — confirm against the datasheet of the specific variant
O: safety exhaust valve YS1 for Y2, Y3	Festo MS6-SV-E (562580)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —
O: redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4	Hydraulic components, nop < 250 000 cycles/year	4	14080	1200	99	e	value from typical values / research — confirm against the datasheet of the specific variant

PL achieved: e — PLr e met

Measures against CCF (to be confirmed by the user): Physical separation of the signal paths of different channels (15); Protection against possible overload (15); Use of well-tried components (5); Protection against CCF due to contamination (mechanics, fluid power systems) and electromagnetic interference (electrical systems) (25); Protection against CCF due to unfavourable environmental conditions (10)

Wiring

- Inputs S2_ChA, S2_ChB: dual-channel into the safety logic, pulsed signals (pulse test) for cross-circuit detection, channel discrepancy monitoring.
- Reset SF_Reset: button outside the hazardous area with a view into it; falling edge evaluation; reset only prepares, the start is given by the standard program.
- KS: two contactors in series controlled by outputs KS1_Q / KS2_Q; NC mirror contacts into EDM KS1_FBK / KS2_FBK (feedback before every start).
- YS: safety valve (supply exhaust / cylinder double valve), coil from output YS1_Q, position feedback YS1_FBK.
- YH: two valves in series, coils YH1_Q / YH2_Q, spool position monitoring YH1_FBK / YH2_FBK.
- Status to the standard PLC: S2_in (auxiliary contact / mirroring from the safety logic) — the standard program reads it only as enable and acknowledgement.

Signal	Direction	Meaning
S2_ChA	input	S2: channel A (NC contact)
S2_ChB	input	S2: channel B (NC contact)
SF_Reset	input	safety functions reset button (outside the hazardous area, with a view into it; falling edge)
KS1_Q	output	KS: safe output 1
KS2_Q	output	KS: safe output 2
KS1_FBK	input	KS: feedback (EDM / position)
KS2_FBK	input	KS: feedback (EDM / position)
YS1_Q	output	YS: safe output 1
YS1_FBK	input	YS: feedback (EDM / position)
YH1_Q	output	YH: safe output 1
YH2_Q	output	YH: safe output 2
YH1_FBK	input	YH: feedback (EDM / position)
YH2_FBK	input	YH: feedback (EDM / position)
S2_in	output	status of S2 to the standard PLC (information signal only, enable / acknowledgement)

Typical components according to sources

- interlocking device types 1–4 according to ISO 14119 (type 1 mech. uncoded, type 2 mech. coded tongue, type 3 non-contact uncoded, type 4 non-contact coded RFID/magnet); ISO 14119:2024 added type 5 = trapped key systems
- contact with direct opening action according to IEC 60947-5-1 Annex K (mechanical switches); type 3/4 as the only switch on the guard must comply with IEC 60947-5-3
- safety module / safety PLC
- contactors with mechanically linked/mirror contacts or STO

Sources

- <https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf>
- https://maxter-automatyka.pl/image/plik_o/960_bezpieczenstwo-maszyn-sick.pdf
- <https://cdn.standards.iteh.ai/samples/75942/ae89318fbad443898f48680f49db9779/ISO-14119-2024.pdf>
- <https://www.reersafety.com/en/academy/industrial-safety-guide/en-iso-14119safety-of-machinery/>
- <https://www.gt-engineering.it/en/technical-standards/en-iso-standards/en-iso-13855/9-interlocking-guards-without-guard-locking/>
- https://literature.rockwellautomation.com/idc/groups/literature/documents/at/safety-at203_-en-p.pdf

SF3 — Interlocking of a movable guard (without guard locking) — S3

Status: function **approved**, design **approved**

SRS field	Value
Description (input → logic → output)	S3 → SIMATIC S7-1200F CPU 1214FC DC/DC/DC → KS, YS, YH
Hazard	Mechanical hazard from moving parts of drives (drawing-in, entanglement, impact, crushing); Crushing and shearing by pneumatic cylinders; residual energy of compressed air; Crushing by hydraulic drive, ejection of pressurised fluid, residual pressure; Crushing and shearing in the press working area (ram / tool stroke); Entering or reaching into the hazardous area during operation
Triggering event	Opening of the guard (leaving the 'closed' position detected by the switch). Inputs: S3 (Right guard closed (NC)).
Reaction	Safety stop (STO, SS1 or SS2) of hazardous movements; start blocked while the guard is open; after closing only manual reset/start (automatic restart only for control guards according to ISO 12100). Outputs: contactors KS1/KS2 with mirror contacts for M1, M2, M3, M4; safety exhaust valve YS1 for Y2, Y3; redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4.
Safe state	drives de-energised (contactors dropped out / STO); pneumatics exhausted; hydraulics depressurised, movement blocked

SRS field	Value
Stop category (IEC 60204-1)	0 — stop category 0: immediate removal of energy (contactors, STO, exhaust)
PLr	e — risk graph S2 / F2 / P2 → e
Operating modes	all operating modes (AUTO and MANUAL) — the standard program does not bypass the protection
Permissible response time	to be completed — at most the time within which a person cannot reach the hazard (ISO 13855)
Demand rate	on average once every 15 min (assumption for nop — verify according to operation)
Priority	stop takes precedence over start; emergency stop over all functions
Reset and restart	Manual reset with button SF_Reset outside the hazardous area (falling edge, monitored signal change); reset does not start the machine — a new start only by the START command of the standard program.
Interface to the standard program	S3_in
Fault reaction	Detected fault (channel discrepancy, short circuit, EDM) → safe state; reset is not possible while the fault persists.
Behaviour on loss of power	Loss of power = safe state (de-energise to trip, outputs de-energised); after power is restored a manual reset is required (manual cold start).
Fault exclusions	none proposed — justify every exclusion (ISO 13849-2); PL e must not rely on fault exclusions alone
Standards	ISO 14119, ISO 13849-1, ISO 13849-2, ISO 13855, ISO 13857, ISO 14120, IEC 60947-5-1, IEC 60947-5-3, IEC 60204-1

Justification of S / F / P (proposal)

- S2: crushing / shearing in the press tool — serious irreversible injury.
- F2: operator interventions (loading, troubleshooting, cleaning) may be more frequent than once per 15 min — adjust according to actual operation.
- P2: the press stroke is fast, a hand in the tool cannot be withdrawn in time.
- Typical according to sources: PLr d; architecture 3 (PL d) typically

Architecture and PL calculation

Category 4, 2 channel(s), EDM yes, manual reset; CCF 70 points (met ≥ 65).

Subsystem	Component	Cat.	nop [1/year]	MTTFd [years]	DC [%]	PL	Notes
I: coded non-contact guard switch (RFID, OSSD)	Schmersal RSS 260 (103003602)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd 6,8E-10 1/h
L: safety logic: Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC	Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC (6ES7214-1AF40-0XB0)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —
O: contactors KS1/KS2 with mirror contacts for M1, M2, M3, M4 — 2 contactors in series, EDM	Siemens SIRIUS 3RT20 / 3RT2	4	14080	973	99	e	MTTFd = B10d / (0.1 · nop) = 1370000 / (0.1 · 14080) = 973 years; value from typical values / research — confirm against the datasheet of the specific variant
O: safety exhaust valve YS1 for Y2, Y3	Festo MS6-SV-E (562580)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —
O: redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4	Hydraulic components, nop < 250 000 cycles/year	4	14080	1200	99	e	value from typical values / research — confirm against the datasheet of the specific variant

PL achieved: e — PLr e met

Measures against CCF (to be confirmed by the user): Physical separation of the signal paths of different channels (15); Protection against possible overload (15); Use of well-trying components (5); Protection against CCF due to contamination (mechanics, fluid power systems) and electromagnetic interference (electrical systems) (25); Protection against CCF due to unfavourable environmental conditions (10)

Wiring

- Inputs S3_ChA, S3_ChB: dual-channel into the safety logic, pulsed signals (pulse test) for cross-circuit detection, channel discrepancy monitoring.
- Reset SF_Reset: button outside the hazardous area with a view into it; falling edge evaluation; reset only prepares, the start is given by the standard program.
- KS: two contactors in series controlled by outputs KS1_Q / KS2_Q; NC mirror contacts into EDM KS1_FBK / KS2_FBK (feedback before every start).
- YS: safety valve (supply exhaust / cylinder double valve), coil from output YS1_Q, position feedback YS1_FBK.
- YH: two valves in series, coils YH1_Q / YH2_Q, spool position monitoring YH1_FBK / YH2_FBK.
- Status to the standard PLC: S3_in (auxiliary contact / mirroring from the safety logic) — the standard program reads it only as enable and acknowledgement.

Signal	Direction	Meaning
S3_ChA	input	S3: channel A (NC contact)
S3_ChB	input	S3: channel B (NC contact)
SF_Reset	input	safety functions reset button (outside the hazardous area, with a view into it; falling edge)
KS1_Q	output	KS: safe output 1
KS2_Q	output	KS: safe output 2
KS1_FBK	input	KS: feedback (EDM / position)
KS2_FBK	input	KS: feedback (EDM / position)
YS1_Q	output	YS: safe output 1
YS1_FBK	input	YS: feedback (EDM / position)
YH1_Q	output	YH: safe output 1
YH2_Q	output	YH: safe output 2
YH1_FBK	input	YH: feedback (EDM / position)
YH2_FBK	input	YH: feedback (EDM / position)
S3_in	output	status of S3 to the standard PLC (information signal only, enable / acknowledgement)

Typical components according to sources

- interlocking device types 1–4 according to ISO 14119 (type 1 mech. uncoded, type 2 mech. coded tongue, type 3 non-contact uncoded, type 4 non-contact coded RFID/magnet); ISO 14119:2024 added type 5 = trapped key systems
- contact with direct opening action according to IEC 60947-5-1 Annex K (mechanical switches); type 3/4 as the only switch on the guard must comply with IEC 60947-5-3
- safety module / safety PLC
- contactors with mechanically linked/mirror contacts or STO

Sources

- <https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf>
- https://maxter-automatyka.pl/image/plik_o/960_bezpieczenstwo-maszyn-sick.pdf
- <https://cdn.standards.itech.ai/samples/75942/ae89318fbad443898f48680f49db9779/ISO-14119-2024.pdf>
- <https://www.reersafety.com/en/academy/industrial-safety-guide/en-iso-14119safety-of-machinery/>
- <https://www.gt-engineering.it/en/technical-standards/en-iso-standards/en-iso-13855/9-interlocking-guards-without-guard-locking/>
- https://literature.rockwellautomation.com/idc/groups/literature/documents/at/safety-at203_-en-p.pdf

SF4 — Guard with guard locking and release of the lock after stopping — S2, S3, H4

Status: function **approved**, design **approved**

SRS field	Value
Description (input → logic → output)	S2, S3, H4 → SIMATIC S7-1200F CPU 1214FC DC/DC/DC → KS, YS, YH
Hazard	Mechanical hazard from moving parts of drives (drawing-in, entanglement, impact, crushing); Crushing and shearing by pneumatic cylinders; residual energy of compressed air; Crushing by hydraulic drive, ejection of pressurised fluid, residual pressure; Crushing and shearing in the press working area (ram / tool stroke); Entering or reaching into the hazardous area during operation
Triggering event	Request to open (button 'unlock') or opening the door while locked is not possible. Inputs: S2 (Left guard closed (NC)), S3 (Right guard closed (NC)), H4 (Electromagnetic guard lock).
Reaction	The guard remains locked as long as the hazardous state persists; unlocking: (a) automatically only on detected standstill (standstill monitor), (b) time-delayed – a timer fault must not shorten the delay, (c) manually – the time from unlocking to guard release longer than the stopping time. The machine can only be started with the guard closed and locked. Outputs: contactors KS1/KS2 with mirror contacts for M1, M2, M3, M4; safety exhaust valve YS1 for Y2, Y3; redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4.
Safe state	drives de-energised (contactors dropped out / STO); pneumatics exhausted; hydraulics depressurised, movement blocked
Stop category (IEC 60204-1)	0 — stop category 0: immediate removal of energy (contactors, STO, exhaust)
PLr	d — risk graph S2 / F1 / P2 → d
Operating modes	all operating modes (AUTO and MANUAL) — the standard program does not bypass the protection
Permissible response time	to be completed — at most the time within which a person cannot reach the hazard (ISO 13855)
Demand rate	on average once every 15 min (assumption for nop — verify according to operation)
Priority	stop takes precedence over start; emergency stop over all functions

SRS field	Value
Reset and restart	Manual reset with button SF_Reset outside the hazardous area (falling edge, monitored signal change); reset does not start the machine — a new start only by the START command of the standard program.
Interface to the standard program	S2_in, S3_in
Fault reaction	Detected fault (channel discrepancy, short circuit, EDM) → safe state; reset is not possible while the fault persists.
Behaviour on loss of power	Loss of power = safe state (de-energise to trip, outputs de-energised); after power is restored a manual reset is required (manual cold start).
Fault exclusions	none proposed — justify every exclusion (ISO 13849-2); PL e must not rely on fault exclusions alone
Standards	ISO 14119, ISO 13849-1, ISO 13849-2, ISO 13855, IEC 60947-5-1, IEC 61800-5-2, IEC 60204-1

Justification of S / F / P (proposal)

- S2: movements of the machine drives can cause serious, normally irreversible injury (fractures, amputation).
- F1: opening the locked guard during overrun is rare.
- P2: the overrun of the movement is longer than the access time — without guard locking the hazard cannot be avoided.
- Typical according to sources: PLr d; architecture 3 (PL d)

Architecture and PL calculation

Category 3, 2 channel(s), EDM yes, manual reset; CCF 70 points (met ≥ 65).

Subsystem	Component	Cat.	nop [1/year]	MTTFd [years]	DC [%]	PL	Notes
I: guard locking device with lock monitoring (principle: locked by spring, unlocked by energy)	Schmersal AZM 201 (103013908)	—	—	—	99	d	PL according to manufacturer's data (with the prescribed wiring); PFHd 1.9E-09 1/h (interlocking); 1.0E-08 1/h (guard locking)
L: safety logic: Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC	Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC (6ES7214-1AF40-0XB0)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —
O: contactors KS1/KS2 with mirror contacts for M1, M2, M3, M4 — 2 contactors in series, EDM	Siemens SIRIUS 3RT20 / 3RT2	3	14080	973	99	d	MTTFd = B10d / (0.1 · nop) = 1370000 / (0.1 · 14080) = 973 years; value from typical values / research — confirm against the datasheet of the specific variant; Channel MTTFd capped at 100 years
O: safety exhaust valve YS1 for Y2, Y3	Festo MS6-SV-E (562580)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —
O: redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4	Hydraulic components, nop < 250 000 cycles/year	3	14080	1200	99	d	value from typical values / research — confirm against the datasheet of the specific variant; Channel MTTFd capped at 100 years

PL achieved: d — PLr d met

Measures against CCF (to be confirmed by the user): Physical separation of the signal paths of different channels (15); Protection against possible overload (15); Use of well-tried components (5); Protection against CCF due to contamination (mechanics, fluid power systems) and electromagnetic interference (electrical systems) (25); Protection against CCF due to unfavourable environmental conditions (10)

Wiring

- Inputs S2_ChA, S2_ChB, S3_ChA, S3_ChB: dual-channel into the safety logic, pulsed signals (pulse test) for cross-circuit detection, channel discrepancy monitoring.
- Reset SF_Reset: button outside the hazardous area with a view into it; falling edge evaluation; reset only prepares, the start is given by the standard program.
- KS: two contactors in series controlled by outputs KS1_Q / KS2_Q; NC mirror contacts into EDM KS1_FBK / KS2_FBK (feedback before every start).
- YS: safety valve (supply exhaust / cylinder double valve), coil from output YS1_Q, position feedback YS1_FBK.
- YH: two valves in series, coils YH1_Q / YH2_Q, spool position monitoring YH1_FBK / YH2_FBK.
- Status to the standard PLC: S2_in, S3_in (auxiliary contact / mirroring from the safety logic) — the standard program reads it only as enable and acknowledgement.

Signal	Direction	Meaning
S2_ChA	input	S2: channel A (NC contact)
S2_ChB	input	S2: channel B (NC contact)
S3_ChA	input	S3: channel A (NC contact)
S3_ChB	input	S3: channel B (NC contact)

Signal	Direction	Meaning
H4_LOCKED	input	H4: locked signal (lock monitoring)
H4_ULC	output	H4: unlock command (only after the hazardous movement has stopped)
SF_Reset	input	safety functions reset button (outside the hazardous area, with a view into it; falling edge)
KS1_Q	output	KS: safe output 1
KS2_Q	output	KS: safe output 2
KS1_FBK	input	KS: feedback (EDM / position)
KS2_FBK	input	KS: feedback (EDM / position)
YS1_Q	output	YS: safe output 1
YS1_FBK	input	YS: feedback (EDM / position)
YH1_Q	output	YH: safe output 1
YH2_Q	output	YH: safe output 2
YH1_FBK	input	YH: feedback (EDM / position)
YH2_FBK	input	YH: feedback (EDM / position)
S2_in	output	status of S2 to the standard PLC (information signal only, enable / acknowledgement)
S3_in	output	status of S3 to the standard PLC (information signal only, enable / acknowledgement)

Typical components according to sources

- electromechanical guard locking device with fail-safe locking element (power-to-unlock)
- standstill monitor / safe speed monitoring or SS1 + timer
- safety module or PLC
- emergency/escape release according to the risk assessment

Sources

- <https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf>
- https://maxter-automatyka.pl/image/plik_o/960_bezpieczenstwo-maszyn-sick.pdf
- <https://machinerysafety101.com/2025/12/17/guard-locking-and-iso-14119/>
- <https://cdn.standards.iteh.ai/samples/75942/ae89318fbad443898f48680f49db9779/ISO-14119-2024.pdf>

SF5 — Light curtain (AOPD) – danger point protection (finger/hand) — S4

Status: function **not approved**, design **not approved**

SRS field	Value
Description (input → logic → output)	S4 → SIMATIC S7-1200F CPU 1214FC DC/DC/DC → KS, YS, YH
Hazard	Mechanical hazard from moving parts of drives (drawing-in, entanglement, impact, crushing); Crushing and shearing by pneumatic cylinders; residual energy of compressed air; Crushing by hydraulic drive, ejection of pressurised fluid, residual pressure; Crushing and shearing in the press working area (ram / tool stroke); Entering or reaching into the hazardous area during operation
Triggering event	Interruption of the protective field (object ≥ resolution d). Inputs: S4 (Light curtain clear).
Reaction	OSSDs switch off → safety stop (STO/SS1) and start interlock; for areas where a person can stand hidden behind the field, manual reset (restart interlock). Outputs: contactors KS1/KS2 with mirror contacts for M1, M2, M3, M4; safety exhaust valve YS1 for Y2, Y3; redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4.
Safe state	drives de-energised (contactors dropped out / STO); pneumatics exhausted; hydraulics depressurised, movement blocked
Stop category (IEC 60204-1)	0 — stop category 0: immediate removal of energy (contactors, STO, exhaust)
PLr	e — risk graph S2 / F2 / P2 → e
Operating modes	all operating modes (AUTO and MANUAL) — the standard program does not bypass the protection
Permissible response time	to be completed — at most the time within which a person cannot reach the hazard (ISO 13855)
Demand rate	on average once every 15 s (assumption for nop — verify according to operation)
Priority	stop takes precedence over start; emergency stop over all functions
Reset and restart	Manual reset with button SF_Reset outside the hazardous area (falling edge, monitored signal change); reset does not start the machine — a new start only by the START command of the standard program.
Interface to the standard program	S4_in
Fault reaction	Detected fault (channel discrepancy, short circuit, EDM) → safe state; reset is not possible while the fault persists.

SRS field	Value
Behaviour on loss of power	Loss of power = safe state (de-energise to trip, outputs de-energised); after power is restored a manual reset is required (manual cold start).
Fault exclusions	none proposed — justify every exclusion (ISO 13849-2); PL e must not rely on fault exclusions alone
Standards	IEC 61496-1, IEC 61496-2, ISO 13855, ISO 13849-1, IEC 62046, IEC 60204-1

Justification of S / F / P (proposal)

- S2: crushing / shearing in the press tool — serious irreversible injury.
- F2: operator interventions (loading, troubleshooting, cleaning) may be more frequent than once per 15 min — adjust according to actual operation.
- P2: the press stroke is fast, a hand in the tool cannot be withdrawn in time.
- Typical according to sources: PLr d or e (manually loaded presses e); architecture AOPD type 4 → cat. 4 / PL e; type 2 → cat. 2 / max. PL c

Architecture and PL calculation

Category 4, 2 channel(s), EDM yes, manual reset; CCF 70 points (met ≥ 65).

Subsystem	Component	Cat.	nop [1/year]	MTTFd [years]	DC [%]	PL	Notes
I: safety light curtain type 4 (IEC 61496), 2 × OSSD	SICK deTec4 Core (1211471 (transmitter C4C-SA06010A10000), 1211472 (receiver C4C-EA06010A10000))	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd 3.7E-09 1/h (TM 20 years)
L: safety logic: Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC	Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC (6ES7214-1AF40-0XB0)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —
O: contactors KS1/KS2 with mirror contacts for M1, M2, M3, M4 — 2 contactors in series, EDM	Siemens SIRIUS 3RT20 / 3RT2	4	844800	16.2	99	—	MTTFd = B10d / (0.1 · nop) = 1370000 / (0.1 · 844800) = 16.2 years; T10D = 1.6 years < mission time 20 years — preventive replacement at the latest after T10D (to be stated in the instructions); value from typical values / research — confirm against the datasheet of the specific variant; combination of category 4, MTTFd "medium" and DCavg "high" is not permitted by the table
O: safety exhaust valve YS1 for Y2, Y3	Festo MS6-SV-E (562580)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —
O: redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4	Hydraulic components, 500 000 ≤ nop < 1 000 000 cycles/year	4	844800	300	99	e	value from typical values / research — confirm against the datasheet of the specific variant

PL achieved: — — PLr e NOT MET

Measures against CCF (to be confirmed by the user): Physical separation of the signal paths of different channels (15); Protection against possible overload (15); Use of well-tried components (5); Protection against CCF due to contamination (mechanics, fluid power systems) and electromagnetic interference (electrical systems) (25); Protection against CCF due to unfavourable environmental conditions (10)

Open points of the design

- Subsystem "contactors KS1/KS2 with mirror contacts for M1, M2, M3, M4 — 2 contactors in series, EDM": PL cannot be determined — combination of category 4, MTTFd "medium" and DCavg "high" is not permitted by the table

Wiring

- Inputs S4_OSSD1, S4_OSSD2: dual-channel into the safety logic, pulsed signals (pulse test) for cross-circuit detection, channel discrepancy monitoring.
- Reset SF_Reset: button outside the hazardous area with a view into it; falling edge evaluation; reset only prepares, the start is given by the standard program.
- KS: two contactors in series controlled by outputs KS1_Q / KS2_Q; NC mirror contacts into EDM KS1_FBK / KS2_FBK (feedback before every start).
- YS: safety valve (supply exhaust / cylinder double valve), coil from output YS1_Q, position feedback YS1_FBK.
- YH: two valves in series, coils YH1_Q / YH2_Q, spool position monitoring YH1_FBK / YH2_FBK.
- Status to the standard PLC: S4_in (auxiliary contact / mirroring from the safety logic) — the standard program reads it only as enable and acknowledgement.

Signal	Direction	Meaning
S4_OSSD1	input	S4: OSSD 1
S4_OSSD2	input	S4: OSSD 2
SF_Reset	input	safety functions reset button (outside the hazardous area, with a view into it; falling edge)

Signal	Direction	Meaning
KS1_Q	output	KS: safe output 1
KS2_Q	output	KS: safe output 2
KS1_FBK	input	KS: feedback (EDM / position)
KS2_FBK	input	KS: feedback (EDM / position)
YS1_Q	output	YS: safe output 1
YS1_FBK	input	YS: feedback (EDM / position)
YH1_Q	output	YH: safe output 1
YH2_Q	output	YH: safe output 2
YH1_FBK	input	YH: feedback (EDM / position)
YH2_FBK	input	YH: feedback (EDM / position)
S4_in	output	status of S4 to the standard PLC (information signal only, enable / acknowledgement)

Minimum distance (ISO 13855:2010)

Formula: $S = 2000 \cdot T + 8 \cdot (d - 14)$

- **missing: measured stopping time of the machine (stopping of the hazardous movement after demand)**
- Δ Press: the C-type standard (EN 692 / EN 693 / ISO 16092) takes precedence — allowance C for $d = 14$ mm is 0 mm; stroke initiation by light curtain permitted.
- Δ Response time of the protective device 15 ms and of the logic 20 ms are a draft — add from the datasheet.

Typical components according to sources

- safety light curtain type 4 with 14 mm (finger) or 30 mm (hand) resolution
- safety module / PLC
- reset button outside the hazardous area with a view
- power elements with feedback

Sources

- https://maxter-automatyka.pl/image/plik_o/960_bezpieczenstwo-maszyn-sick.pdf
- <https://www.hse.gov.uk/pubns/priced/hsg180.pdf>
- <https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf>
- <https://publikationen.dguv.de/widgets/pdf/download/article/5091>
- <https://www.keyence.eu/ss/products/safetyknowledge/caution/>
- <https://www.gt-engineering.it/en/technical-standards/en-iso-standards/en-iso-13855/6-detection-zone-orthogonal-to-the-direction-of-approach/>

SF6 — Two-hand control — S5, S6

Status: function **not approved**, design **not approved**

SRS field	Value
Description (input → logic → output)	S5, S6 → SIMATIC S7-1200F CPU 1214FC DC/DC/DC → YH
Hazard	Crushing by hydraulic drive, ejection of pressurised fluid, residual pressure; Crushing and shearing in the press working area (ram / tool stroke); Entering or reaching into the hazardous area during operation
Triggering event	Release of either of the two actuators (or non-synchronous actuation for type III). Inputs: S5 (Two-hand start — button A), S6 (Two-hand start — button B).
Reaction	Output signal only while both actuators are held simultaneously; releasing one terminates the hazardous movement; a new cycle only after releasing both and pressing again (type II/III); type III additionally synchronous actuation ≤ 0.5 s. Outputs: redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4.
Safe state	hydraulics depressurised, movement blocked
Stop category (IEC 60204-1)	0 — stop category 0: immediate removal of energy (contactors, STO, exhaust)
PLr	e — risk graph S2 / F2 / P2 → e; normative minimum e: ISO 13851:2019: two-hand control type IIIC (presses) = PL e, category 4.
Operating modes	all operating modes (AUTO and MANUAL) — the standard program does not bypass the protection
Permissible response time	to be completed — at most the time within which a person cannot reach the hazard (ISO 13855)
Demand rate	on average once every 15 s (assumption for nop — verify according to operation)
Priority	stop takes precedence over start; emergency stop over all functions

SRS field	Value
Reset and restart	Manual reset with button SF_Reset outside the hazardous area (falling edge, monitored signal change); reset does not start the machine — a new start only by the START command of the standard program.
Interface to the standard program	S5_in, S6_in
Fault reaction	Detected fault (channel discrepancy, short circuit, EDM) → safe state; reset is not possible while the fault persists.
Behaviour on loss of power	Loss of power = safe state (de-energise to trip, outputs de-energised); after power is restored a manual reset is required (manual cold start).
Fault exclusions	none proposed — justify every exclusion (ISO 13849-2); PL e must not rely on fault exclusions alone
Standards	ISO 13851, ISO 13855, ISO 13849-1, IEC 60947-5-1, EN 692 / ISO 16092 (lisy)

Justification of S / F / P (proposal)

- S2: crushing / shearing in the press tool — serious irreversible injury.
- F2: the operator reaches into the tool in every cycle.
- P2: the press stroke is fast, a hand in the tool cannot be withdrawn in time.
- Typical according to sources: PLr e (type IIIC) for presses; otherwise according to type; architecture 4 (IIIC) / 3 (IIIB, II) / 1 (I, IIIA)

Architecture and PL calculation

Category 4, 2 channel(s), EDM yes, manual reset; CCF 70 points (met ≥ 65).

Subsystem	Component	Cat.	nop [1/year]	MTTFd [years]	DC [%]	PL	Notes
I: two-hand station: 2 buttons, each NO + NC contact (antivalent)	Siemens SIRIUS ACT 3SU1 – two-hand control station (3SU1853-3NB00-1AA1)	4	844800	591.9	99	e	MTTFd = B10d / (0.1 · nop) = 50000000 / (0.1 · 844800) = 591.9 years; value from typical values / research — confirm against the datasheet of the specific variant
L: safety logic: Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC	Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC (6ES7214-1AF40-0XB0)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —
O: redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4	Hydraulic components, 500 000 ≤ nop < 1 000 000 cycles/year	4	844800	300	99	e	value from typical values / research — confirm against the datasheet of the specific variant

PL achieved: e — PLr e met

Measures against CCF (to be confirmed by the user): Physical separation of the signal paths of different channels (15); Protection against possible overload (15); Use of well-tried components (5); Protection against CCF due to contamination (mechanics, fluid power systems) and electromagnetic interference (electrical systems) (25); Protection against CCF due to unfavourable environmental conditions (10)

Wiring

- Inputs S5_NO, S5_NC, S6_NO, S6_NC: dual-channel into the safety logic, pulsed signals (pulse test) for cross-circuit detection, channel discrepancy monitoring.
- Reset SF_Reset: button outside the hazardous area with a view into it; falling edge evaluation; reset only prepares, the start is given by the standard program.
- YH: two valves in series, coils YH1_Q / YH2_Q, spool position monitoring YH1_FBK / YH2_FBK.
- Status to the standard PLC: S5_in, S6_in (auxiliary contact / mirroring from the safety logic) — the standard program reads it only as enable and acknowledgement.

Signal	Direction	Meaning
S5_NO	input	S5: button, NO contact
S5_NC	input	S5: button, NC contact
S6_NO	input	S6: button, NO contact
S6_NC	input	S6: button, NC contact
SF_Reset	input	safety functions reset button (outside the hazardous area, with a view into it; falling edge)
YH1_Q	output	YH: safe output 1
YH2_Q	output	YH: safe output 2
YH1_FBK	input	YH: feedback (EDM / position)
YH2_FBK	input	YH: feedback (EDM / position)
S5_in	output	status of S5 to the standard PLC (information signal only, enable / acknowledgement)
S6_in	output	status of S6 to the standard PLC (information signal only, enable / acknowledgement)

Minimum distance (ISO 13855:2010)

Formula: $S = 1600 \cdot T + 250$

- **missing: measured stopping time of the machine (stopping of the hazardous movement after demand)**
- Δ Response time of the protective device 0 ms and of the logic 20 ms are a draft — add from the datasheet.

Typical components according to sources

- 2 actuators according to IEC 60947-5-1 with shrouding against defeating (elbows, one hand)
- two-hand control logic type IIIC / safety PLC
- contactors / valves with feedback

Sources

- <https://cdn.standards.iteh.ai/samples/70295/48c735b3148b4aa4943c9f1095f6541a/ISO-13851-2019.pdf>
- <https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf>
- https://maxter-automatyka.pl/image/plik_o/960_bezpieczenstwo-maszyn-sick.pdf
- https://www.safe-machines-at-work.org/fileadmin/user_upload/Two-Hand-Controls_08_2020.pdf
- https://literature.rockwellautomation.com/idc/groups/literature/documents/at/safety-at071_-en-e.pdf
- <https://www.gt-engineering.it/en/technical-standards/en-iso-standards/en-iso-13855/8-two-hand-control-devices/>

SF7 — Prevention of unexpected start-up (isolation and energy dissipation, LOTO, start interlock)

Status: function **not approved**, design **not approved**

SRS field	Value
Description (input → logic → output)	— → SIMATIC S7-1200F CPU 1214FC DC/DC/DC → KS, YS, YH
Hazard	Unexpected start-up after intervention, reset or restoration of energy
Triggering event	Intervention of a person in the hazardous area (guard opened, maintenance mode, isolator in position 0).
Reaction	The machine remains safely stopped while a person is in the hazardous area; neither restoration of power nor reset may start the machine. Isolation and energy dissipation procedure: isolate from all sources, lock the isolators in the 'isolated' position, dissipate or restrain stored energy. Outputs: contactors KS1/KS2 with mirror contacts for M1, M2, M3, M4; safety exhaust valve YS1 for Y2, Y3; redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4.
Safe state	drives de-energised (contactors dropped out / STO); pneumatics exhausted; hydraulics depressurised, movement blocked
Stop category (IEC 60204-1)	0 — stop category 0: immediate removal of energy (contactors, STO, exhaust)
PLr	d — risk graph S2 / F1 / P2 → d
Operating modes	all operating modes (AUTO and MANUAL) — the standard program does not bypass the protection
Permissible response time	to be completed — at most the time within which a person cannot reach the hazard (ISO 13855)
Demand rate	on average once every 1 h (assumption for nop — verify according to operation)
Priority	stop takes precedence over start; emergency stop over all functions
Reset and restart	Manual reset with button SF_Reset outside the hazardous area (falling edge, monitored signal change); reset does not start the machine — a new start only by the START command of the standard program.
Interface to the standard program	no status signal
Fault reaction	Detected fault (channel discrepancy, short circuit, EDM) → safe state; reset is not possible while the fault persists.
Behaviour on loss of power	Loss of power = safe state (de-energise to trip, outputs de-energised); after power is restored a manual reset is required (manual cold start).
Fault exclusions	none proposed — justify every exclusion (ISO 13849-2); PL e must not rely on fault exclusions alone
Standards	ISO 14118, IEC 60204-1, ISO 4413, ISO 4414, ISO 12100, ISO 13849-1

Justification of S / F / P (proposal)

- S2: movements of the machine drives can cause serious, normally irreversible injury (fractures, amputation).
- F1: interventions followed by a risk of unexpected start-up are rare.
- P2: the operator does not anticipate an unexpected start-up.
- Typical according to sources: PLr according to the function it protects (typically d); architecture according to PLr

Architecture and PL calculation

Category 3, 2 channel(s), EDM yes, manual reset; CCF 70 points (met ≥ 65).

Subsystem	Component	Cat.	nop [1/year]	MTTFd [years]	DC [%]	PL	Notes
I: reset button (NO + NC contact, monitored edge)	Buttons / enabling switches (IEC 60947-5-8)	2	3520	284.1	90	d	$MTTFd = B10d / (0.1 \cdot nop) = 100000 / (0.1 \cdot 3520) = 284.1$ years; value from typical values / research — confirm against the datasheet of the specific variant; Channel MTTFd capped at 100 years
L: safety logic: Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC	Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC (6ES7214-1AF40-0XB0)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —
O: contactors KS1/KS2 with mirror contacts for M1, M2, M3, M4 — 2 contactors in series, EDM	Siemens SIRIUS 3RT20 / 3RT2	3	3520	3892	99	d	$MTTFd = B10d / (0.1 \cdot nop) = 1370000 / (0.1 \cdot 3520) = 3892$ years; value from typical values / research — confirm against the datasheet of the specific variant; Channel MTTFd capped at 100 years
O: safety exhaust valve YS1 for Y2, Y3	Festo MS6-SV-E (562580)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —
O: redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4	Hydraulic components, nop < 250 000 cycles/year	3	3520	1200	99	d	value from typical values / research — confirm against the datasheet of the specific variant; Channel MTTFd capped at 100 years

PL achieved: d — PLr d met

Measures against CCF (to be confirmed by the user): Physical separation of the signal paths of different channels (15); Protection against possible overload (15); Use of well-tried components (5); Protection against CCF due to contamination (mechanics, fluid power systems) and electromagnetic interference (electrical systems) (25); Protection against CCF due to unfavourable environmental conditions (10)

Wiring

- Reset SF_Reset: button outside the hazardous area with a view into it; falling edge evaluation; reset only prepares, the start is given by the standard program.
- KS: two contactors in series controlled by outputs KS1_Q / KS2_Q; NC mirror contacts into EDM KS1_FBK / KS2_FBK (feedback before every start).
- YS: safety valve (supply exhaust / cylinder double valve), coil from output YS1_Q, position feedback YS1_FBK.
- YH: two valves in series, coils YH1_Q / YH2_Q, spool position monitoring YH1_FBK / YH2_FBK.

Signal	Direction	Meaning
SF_Reset	input	safety functions reset button (outside the hazardous area, with a view into it; falling edge)
KS1_Q	output	KS: safe output 1
KS2_Q	output	KS: safe output 2
KS1_FBK	input	KS: feedback (EDM / position)
KS2_FBK	input	KS: feedback (EDM / position)
YS1_Q	output	YS: safe output 1
YS1_FBK	input	YS: feedback (EDM / position)
YH1_Q	output	YH: safe output 1
YH2_Q	output	YH: safe output 2
YH1_FBK	input	YH: feedback (EDM / position)
YH2_FBK	input	YH: feedback (EDM / position)

Typical components according to sources

- lockable main switch (isolator) according to IEC 60204-1
- lockable shut-off valve with exhaust (pneumatics) – ISO 4414
- discharging / isolation of the hydraulic accumulator – ISO 4413
- drive STO
- blocking/supporting of vertical axes (see vertical_axis_holding)

Sources

- <https://www.gt-engineering.it/en/technical-standards/en-iso-standards/iso-14118/>
- <https://publikationen.dguv.de/widgets/pdf/download/article/5091>
- https://maxter-automatyka.pl/image/plik_o/960_bezpieczenstwo-maszyn-sick.pdf
- <https://standards.iteh.ai/catalog/standards/cen/365f07d1-cb0f-4414-9fe9-240ac4c21109/en-iso-4414-2010>
- <https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf>

SF8 — Safe pneumatic exhaust (SDE) and blocking of pneumatic movements (SSC)

Status: function **not approved**, design **not approved**

SRS field	Value
Description (input → logic → output)	— → SIMATIC S7-1200F CPU 1214FC DC/DC/DC → YS
Hazard	Crushing and shearing by pneumatic cylinders; residual energy of compressed air
Triggering event	Safety function demand (E-stop, opening of the guard). Required by: SF1, SF2, SF3, SF4, SF5, SF7.
Reaction	SDE: safe exhaust (for hydraulics pressure relief); SSC: drive stopped, supply/exhaust of at least one chamber closed; STO (fluid): no energy supplied to the drive, energy in the chambers is released. If trapped air presents a further hazard, additional measures are required. Outputs: safety exhaust valve YS1 for Y2, Y3.
Safe state	pneumatics exhausted
Stop category (IEC 60204-1)	0 — stop category 0: immediate removal of energy (contactors, STO, exhaust)
PLr	e — risk graph S2 / F2 / P2 → e
Operating modes	all operating modes (AUTO and MANUAL) — the standard program does not bypass the protection
Permissible response time	to be completed — at most the time within which a person cannot reach the hazard (ISO 13855)
Demand rate	on average once every 15 s (assumption for nop — verify according to operation)
Priority	stop takes precedence over start; emergency stop over all functions
Reset and restart	Manual reset with button SF_Reset outside the hazardous area (falling edge, monitored signal change); reset does not start the machine — a new start only by the START command of the standard program.
Interface to the standard program	no status signal
Fault reaction	Detected fault (channel discrepancy, short circuit, EDM) → safe state; reset is not possible while the fault persists.
Behaviour on loss of power	Loss of power = safe state (de-energise to trip, outputs de-energised); after power is restored a manual reset is required (manual cold start).
Fault exclusions	none proposed — justify every exclusion (ISO 13849-2); PL e must not rely on fault exclusions alone
Standards	ISO 4414, ISO 13849-1, ISO 13849-2, ISO 14118, VDMA 24584

Justification of S / F / P (proposal)

- taken over from SF2 (highest PLr of the functions requiring this output)
- F2: operator interventions (loading, troubleshooting, cleaning) may be more frequent than once per 15 min — adjust according to actual operation.
- P2: the press stroke is fast, a hand in the tool cannot be withdrawn in time.
- Typical according to sources: PLr d (e for presses / high risk); architecture 3 or 4

Architecture and PL calculation

Category 4, 2 channel(s), EDM yes, manual reset; CCF 70 points (met ≥ 65).

Subsystem	Component	Cat.	nop [1/year]	MTTFd [years]	DC [%]	PL	Notes
L: safety logic: Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC	Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC (6ES7214-1AF40-0XB0)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —
O: safety exhaust valve YS1 for Y2, Y3	Festo MS6-SV-E (562580)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —

PL achieved: e — PLr e met

Measures against CCF (to be confirmed by the user): Physical separation of the signal paths of different channels (15); Protection against possible overload (15); Use of well-tried components (5); Protection against CCF due to contamination (mechanics, fluid power systems) and electromagnetic interference (electrical systems) (25); Protection against CCF due to unfavourable environmental conditions (10)

Wiring

- Reset SF_Reset: button outside the hazardous area with a view into it; falling edge evaluation; reset only prepares, the start is given by the standard program.
- YS: safety valve (supply exhaust / cylinder double valve), coil from output YS1_Q, position feedback YS1_FBK.

Signal	Direction	Meaning
SF_Reset	input	safety functions reset button (outside the hazardous area, with a view into it; falling edge)
YS1_Q	output	YS: safe output 1
YS1_FBK	input	YS: feedback (EDM / position)

Typical components according to sources

- safety exhaust valve (redundant, monitored)
- directional valve with closed centre position and spring centring
- pressure switch
- piston rod brake/clamp (vertical axes)
- lockable shut-off valve with exhaust (ISO 4414)

Sources

- <https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf>
- <https://publikationen.dguv.de/widgets/pdf/download/article/5091>
- https://literature.rockwellautomation.com/idc/groups/literature/documents/at/safety-at128_-en-p.pdf
- <https://docs.rs-online.com/34ef/A700000007850995.pdf>
- <https://standards.iteh.ai/catalog/standards/cen/365f07d1-cb0f-4414-9fe9-240ac4c21109/en-iso-4414-2010>

SF9 — Safe stop / isolation of hydraulics (redundant valves with position monitoring)

Status: function **not approved**, design **not approved**

SRS field	Value
Description (input → logic → output)	— → SIMATIC S7-1200F CPU 1214FC DC/DC/DC → YH
Hazard	Crushing by hydraulic drive, ejection of pressurised fluid, residual pressure
Triggering event	Safety function demand (protective device, E-stop). Required by: SF1, SF2, SF3, SF4, SF5, SF6, SF7.
Reaction	Stopping the hazardous movement and preventing start-up from standstill with two independent valves; when a valve fault is detected, block further hazardous movement; if necessary pressure relief (SDE) and isolation/discharge of the accumulator. Outputs: redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4.
Safe state	hydraulics depressurised, movement blocked
Stop category (IEC 60204-1)	0 — stop category 0: immediate removal of energy (contactors, STO, exhaust)
PLr	e — risk graph S2 / F2 / P2 → e
Operating modes	all operating modes (AUTO and MANUAL) — the standard program does not bypass the protection
Permissible response time	to be completed — at most the time within which a person cannot reach the hazard (ISO 13855)
Demand rate	on average once every 15 s (assumption for nop — verify according to operation)
Priority	stop takes precedence over start; emergency stop over all functions
Reset and restart	Manual reset with button SF_Reset outside the hazardous area (falling edge, monitored signal change); reset does not start the machine — a new start only by the START command of the standard program.
Interface to the standard program	no status signal
Fault reaction	Detected fault (channel discrepancy, short circuit, EDM) → safe state; reset is not possible while the fault persists.
Behaviour on loss of power	Loss of power = safe state (de-energise to trip, outputs de-energised); after power is restored a manual reset is required (manual cold start).
Fault exclusions	none proposed — justify every exclusion (ISO 13849-2); PL e must not rely on fault exclusions alone
Standards	ISO 4413, ISO 13849-1, ISO 13849-2, ISO 16092-3 (hydraulic presses), VDMA 24584

Justification of S / F / P (proposal)

- taken over from SF2 (highest PLr of the functions requiring this output)
- F2: operator interventions (loading, troubleshooting, cleaning) may be more frequent than once per 15 min — adjust according to actual operation.
- P2: the press stroke is fast, a hand in the tool cannot be withdrawn in time.
- Typical according to sources: PLr d to e (presses e); architecture 3 or 4

Architecture and PL calculation

Category 4, 2 channel(s), EDM yes, manual reset; CCF 70 points (met ≥ 65).

Subsystem	Component	Cat.	nop [1/year]	MTTFd [years]	DC [%]	PL	Notes
L: safety logic: Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC	Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC (6ES7214-1AF40-0XB0)	—	—	—	99	e	PL according to manufacturer's data (with the prescribed wiring); PFHd —

Subsystem	Component	Cat.	nop [1/year]	MTTFd [years]	DC [%]	PL	Notes
O: redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4	Hydraulic components, 500 000 ≤ nop < 1 000 000 cycles/year	4	844800	300	99	e	value from typical values / research — confirm against the datasheet of the specific variant

PL achieved: e — PLr e met

Measures against CCF (to be confirmed by the user): Physical separation of the signal paths of different channels (15); Protection against possible overload (15); Use of well-tried components (5); Protection against CCF due to contamination (mechanics, fluid power systems) and electromagnetic interference (electrical systems) (25); Protection against CCF due to unfavourable environmental conditions (10)

Wiring

- Reset SF_Reset: button outside the hazardous area with a view into it; falling edge evaluation; reset only prepares, the start is given by the standard program.
- YH: two valves in series, coils YH1_Q / YH2_Q, spool position monitoring YH1_FBK / YH2_FBK.

Signal	Direction	Meaning
SF_Reset	input	safety functions reset button (outside the hazardous area, with a view into it; falling edge)
YH1_Q	output	YH: safe output 1
YH2_Q	output	YH: safe output 2
YH1_FBK	input	YH: feedback (EDM / position)
YH2_FBK	input	YH: feedback (EDM / position)

Typical components according to sources

- redundant directional / logic valves with position sensing
- pressure relief valve (pressure limiting)
- filtration
- accumulator safety block (isolation and discharge)

Sources

- <https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf>
- <https://publikationen.dguv.de/widgets/pdf/download/article/5091>

SF10 — Pressure limitation (relief valve) and pressure monitoring

Status: function **not approved**, design **not approved**

SRS field	Value
Description (input → logic → output)	pressure relief valve set to the maximum permissible pressure (mechanical principle, ISO 4413 / pressure equipment) → SIMATIC S7-1200F CPU 1214FC DC/DC/DC → YH
Hazard	Crushing by hydraulic drive, ejection of pressurised fluid, residual pressure; Overpressure in a pressure system (rupture, escape of medium)
Triggering event	Pressure above the set limit.
Reaction	A relief valve (hydraulics) / pressure regulator with secondary exhaust (pneumatics) prevents pressure rising above the set level – mechanically, independently of the control system. Pressure monitoring by a pressure switch is used in the IFA examples as valve diagnostics. Outputs: redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4.
Safe state	hydraulics depressurised, movement blocked
Stop category (IEC 60204-1)	0 — stop category 0: immediate removal of energy (contactors, STO, exhaust)
PLr	— — risk graph S2 / F1 / P2 → d
Operating modes	all operating modes (AUTO and MANUAL) — the standard program does not bypass the protection
Permissible response time	to be completed — at most the time within which a person cannot reach the hazard (ISO 13855)
Demand rate	on average once every 1 d (assumption for nop — verify according to operation)
Priority	stop takes precedence over start; emergency stop over all functions
Reset and restart	Manual reset with button SF_Reset outside the hazardous area (falling edge, monitored signal change); reset does not start the machine — a new start only by the START command of the standard program.
Interface to the standard program	no status signal
Fault reaction	Detected fault (channel discrepancy, short circuit, EDM) → safe state; reset is not possible while the fault persists.
Behaviour on loss of power	Loss of power = safe state (de-energise to trip, outputs de-energised); after power is restored a manual reset is required (manual cold start).

SRS field	Value
Fault exclusions	none proposed — justify every exclusion (ISO 13849-2); PL e must not rely on fault exclusions alone
Standards	ISO 4413, ISO 4414, ISO 13849-2 (Annex B, C), Directive 2014/68/EU (pressure equipment) – not verified in this research

Justification of S / F / P (proposal)

- S2: rupture of the pressure system.
- F1: demand is rare and exposure time short (less than once per 15 min, cumulatively below 1/20 of the operating time).
- P2: avoiding the hazard is scarcely possible (fast movement, confined space, hazard may not be visible).
- Typical according to sources: PLr not applicable (mechanical principle) / not verified; architecture B (basic principle)

Architecture and PL calculation

Category B, 1 channel(s), EDM yes, manual reset; CCF 70 points (met $\geq$ 65).

Subsystem	Component	Cat.	nop [1/year]	MTTFd [years]	DC [%]	PL	Notes
O: redundant hydraulic valves YH1/YH2 with position monitoring for Y1, Y4, Y6, M4	Hydraulic components, nop < 250 000 cycles/year	B	147	1200	0	b	value from typical values / research — confirm against the datasheet of the specific variant; Channel MTTFd capped at 100 years; category B: at most PL b

PL achieved: — — PLr — NOT MET

Measures against CCF (to be confirmed by the user): Physical separation of the signal paths of different channels (15); Protection against possible overload (15); Use of well-tried components (5); Protection against CCF due to contamination (mechanics, fluid power systems) and electromagnetic interference (electrical systems) (25); Protection against CCF due to unfavourable environmental conditions (10)

Open points of the design

- Pressure is protected by a pressure relief valve (mechanical basic principle) — PL is not calculated; the pressure switch and alarm are supplementary. Add the valve type and setting.
- Missing: pressure relief valve set to the maximum permissible pressure (mechanical principle, ISO 4413 / pressure equipment)

Wiring

- YH: two valves in series, coils YH1_Q / YH2_Q, spool position monitoring YH1_FBK / YH2_FBK.

Signal	Direction	Meaning
YH1_Q	output	YH: safe output 1
YH2_Q	output	YH: safe output 2
YH1_FBK	input	YH: feedback (EDM / position)
YH2_FBK	input	YH: feedback (EDM / position)

Typical components according to sources

- relief (overpressure) valve
- pressure switch / transducer (diagnostics)
- fluid filtration

Sources

- <https://publikationen.dguv.de/widgets/pdf/download/article/5091>
- <https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf>
- <https://standards.iteh.ai/catalog/standards/cen/365f07d1-cb0f-4414-9fe9-240ac4c21109/en-iso-4414-2010>

4. What the responsible person approves

1. Limits of the machine, intended use and reasonably foreseeable misuse.
2. Hazard identification, risk estimation and evaluation, and choice of measures in the order design → protective measures → information; the application does not decide that the risk is acceptable.
3. List of safety functions and their assignment to hazards.
4. PLr of each function (S, F, P and probability of occurrence) or adoption from a type-C standard; reduction of PLr only with written justification.
5. The complete specification (SRS) and its review before design.
6. Architecture, category and component selection including every fault exclusion with justification (PL e must not rely on fault exclusions alone).
7. Stopping time and minimum distances — only from measured values; the formula and the ISO 13855 edition are stated.
8. Validation plan, FAT report and validation report — results are recorded by a person according to the actual test; simulation verifies the design, not the machine.
9. Safety program / configuration of the safety logic (item safety:program).
10. Residual risks, texts for the instruction manual and procedures for periodic testing of safety functions.

11. Technical documentation, EU declaration of conformity and conformity assessment procedure (Regulation (EU) 2023/1230 from 20 Jan 2027).
12. Cybersecurity: protection of safety software and parameters against unauthorised modification.
13. Assessment of every modification: scope of repeated verification and validation, possibly a substantial modification of the machine.

5. Conflicts between sources (research)

Topic	Resolution	Sources
Reduction of PLr by one level for a low probability of occurrence	The IFA/DGUV interpretation takes precedence (the authors are members of the standardisation group).	https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf https://publikationen.dguv.de/widgets/pdf/download/article/4894
Probability of occurrence in the 2023 edition	unresolved — the user decides	https://publikationen.dguv.de/widgets/pdf/download/article/4894 https://www.linkedin.com/pulse/changes-iso-138492023-martin-kidman-ph-d-
Factors of parameter P	For 2023 use the table of five factors (A/B/C).	https://www.pilz.com/en-GB/support/law-standards-norms/functional-safety/en-iso-13849-1 https://publikationen.dguv.de/widgets/pdf/download/article/4894
Criteria of parameter F in the 2023 edition	unresolved — the user decides	https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf https://www.interventionprevention.com/en/iso-13849-1-2023/
PL ↔ PFH table on the CTB website	unresolved — the user decides	https://www.ctb.co.at/en/knowledge/safety-performance-level-sil-iec-13849-62061/ https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf
PL ↔ SIL correspondence	unresolved — the user decides	https://literature.rockwellautomation.com/idc/groups/literature/documents/rm/safebk-rm002_-en-p.pdf https://www.ctb.co.at/en/knowledge/safety-performance-level-sil-iec-13849-62061/
IEC 62061 matrix: 2005 vs. 2021	unresolved — the user decides	https://literature.rockwellautomation.com/idc/groups/literature/documents/rm/safebk-rm002_-en-p.pdf https://www.linkedin.com/pulse/new-risk-graph-en-iec-620612021-martin-kidman-ph-d-
Validity of editions	unresolved — the user decides	https://www.pilz.com/en-GB/support/law-standards-norms/functional-safety/en-iso-13849-1
B10D of a contactor at rated load	Use 1 300 000 (2015); value in ISO 13849-1:2023 table C.1 not verified.	https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf https://assets.new.siemens.com/siemens/assets/api/uuid:23ce849b-fca0-4402-b88b-91adef4139e8/hilfetexte-iso-en-211221.pdf
Push buttons and position switches without fault exclusion of direct opening action	unresolved — the user decides	https://assets.new.siemens.com/siemens/assets/api/uuid:23ce849b-fca0-4402-b88b-91adef4139e8/hilfetexte-iso-en-211221.pdf https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf
Hydraulic components	unresolved — the user decides	https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf
Conversion B10 → B10D	unresolved — the user decides	https://publikationen.dguv.de/widgets/pdf/download/article/4894
DCavg for categories 2 and 3	unresolved — the user decides	https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf https://assets.new.siemens.com/siemens/assets/api/uuid:23ce849b-fca0-4402-b88b-91adef4139e8/hilfetexte-iso-en-211221.pdf
Table 6 (bar chart in the table)	unresolved — the user decides	https://publikationen.dguv.de/widgets/pdf/download/article/4894 https://assets.new.siemens.com/siemens/assets/api/uuid:23ce849b-fca0-4402-b88b-91adef4139e8/hilfetexte-iso-en-211221.pdf
Numbering of the DC annex in the 2023 edition	unresolved — the user decides	https://publikationen.dguv.de/widgets/pdf/download/article/4894
Stop category for emergency stop	Category 0 or 1 applies.	https://literature.rockwellautomation.com/idc/groups/literature/documents/rm/safebk-rm002_-en-p.pdf https://machinerysafety101.com/2010/09/27/emergency-stop-categories/

Topic	Resolution	Sources
ISO 13855: allowance for resolution 40–70 mm	Difference between editions. The 2024 values come from only one freely available interpretation (GT Engineering) – verify in the standard before use.	https://maxter-automatyka.pl/image/plik_o/960_bezpieczenstwo-maszyn-sick.pdf https://www.keyence.eu/ss/products/safetyknowledge/caution/
ISO 13855: parallel approach	Difference between editions; 2024 is more conservative (higher S).	https://maxter-automatyka.pl/image/plik_o/960_bezpieczenstwo-maszyn-sick.pdf https://www.keyence.eu/ss/products/safetyknowledge/caution/
ISO 13855: two-hand control	ISSA leaflet 2020 and SICK 2015 describe the 2010 state; the value 550 mm only from GT Engineering – verify.	https://maxter-automatyka.pl/image/plik_o/960_bezpieczenstwo-maszyn-sick.pdf https://www.safe-machines-at-work.org/fileadmin/user_upload/Two-Hand-Controls_08_2020.pdf
ISO 13855: reaching under a multi-beam light grid	Unclear whether these are different rules (fixed beam height × calculated reach-under). Verify in the standard.	https://mpemagazine.co.uk/2025/08/01/leuze-on-iso-138552024-new-guidelines-for-safer-sensor-positioning-in-industrial-automation/ https://git-sicherheit.de/de/topstories/zur-aktualisierten-version-der-iso-13855-zur-anordnung-von-schutzeinrichtungen-an-maschinen
Harmonisation of EN ISO 13855:2024	Check the current list of harmonised standards in the Official Journal of the EU.	https://git-sicherheit.de/de/topstories/zur-aktualisierten-version-der-iso-13855-zur-anordnung-von-schutzeinrichtungen-an-maschinen
Two-hand control – PL by type	The table of the standard applies (ISO preview); the column assignment was read from a PDF extract – verify visually in the standard.	https://cdn.standards.iteh.ai/samples/70295/48c735b3148b4aa4943c9f1095f6541a/ISO-13851-2019.pdf https://industrialmonitordirect.com/blogs/knowledgebase/implementing-two-hand-control-safety-per-iso-13851
Emergency stop – PLr	No contradiction – c is the minimum of the standard, d is the typical choice in the examples; higher PLr according to the risk assessment.	https://www.idec.com/en-eu/solutions/safety/law/iso-iec/iso13850 https://machinerysafety101.com/2026/05/18/iso-13850-emergency-stop-requirements/
Single-action electromechanical devices (E-stop, guard switch) and PL e	Consistent: the admissibility of fault exclusion for the given element is decisive.	https://literature.rockwellautomation.com/idc/groups/literature/documents/at/safety-at203_-en-p.pdf https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf
Manual reset – edge	Not a contradiction – IFA specifies the implementation; the time window is a parameter of the example, not of the standard.	https://maxter-automatyka.pl/image/plik_o/960_bezpieczenstwo-maszyn-sick.pdf https://machinerysafety101.com/2021/05/19/understanding-safety-functions-manual-reset/

Topic	Resolution	Sources
Muting – status of the standard and times	Verify 4 s in IEC 62046:2018; do not state the override time as a normative value.	https://www.sick.com/media/docs/9/99/499/whitepaper_protecting_automated_loading_and_unloading_points_en_im0072499.pdf https://cdn.standards.iteh.ai/samples/20537/2ef12b276a9c45c1b70d3caab93b1e71/IEC-62046-2018.pdf
SICK muting whitepaper – obvious error	Use $d \leq 40$ mm.	https://www.sick.com/media/docs/9/99/499/whitepaper_protecting_automated_loading_and_unloading_points_en_im0072499.pdf https://maxter-automatyka.pl/image/plik_o/960_bezpieczenstwo-maszyn-sick.pdf
Periodic inspection of ESPE	British practice (PUWER), not an EU standard; in the Czech Republic the intervals are set by the manufacturer/risk assessment.	https://www.hse.gov.uk/pubns/priced/hsg180.pdf
IFA report version	For new projects prefer IFA 1/2025; example numbers differ between editions (e.g. muting ex. 22 → ex. 17).	https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf https://publikationen.dguv.de/widgets/pdf/download/article/5091
Guard locking principle	Sources agree.	https://maxter-automatyka.pl/image/plik_o/960_bezpieczenstwo-maszyn-sick.pdf https://machinerysafety101.com/2025/12/17/guard-locking-and-iso-14119/

6. Sources

- <https://www.dguv.de/medien/ifa/en/pub/rep/pdf/reports-2019/report0217e/rep0217e.pdf>
- https://literature.rockwellautomation.com/idc/groups/literature/documents/rm/safebk-rm002_-en-p.pdf
- <https://assets.new.siemens.com/siemens/assets/api/uuid:23ce849b-fca0-4402-b88b-91a9ef4139e8/hilfetexte-iso-en-211221.pdf>
- <https://publikationen.dguv.de/widgets/pdf/download/article/4894>
- <https://www.se.com/us/en/faqs/FA122781/>
- <https://machinerysafety101.com/2010/09/27/emergency-stop-categories/>
- https://maxter-automatyka.pl/image/plik_o/960_bezpieczenstwo-maszyn-sick.pdf
- <https://www.keyence.eu/ss/products/safetyknowledge/caution/>
- <https://www.reersafety.com/en/academy/industrial-safety-guide/determination-of-the-safety-distance/>
- <https://git-sicherheit.de/de/topstories/zur-aktualisierten-version-der-iso-13855-zur-anordnung-von-schutzeinrichtungen-an-maschinen>
- <https://www.machinebuilding.net/iso-138552024--new-guidelines-for-safer-sensor-positioning-in-industrial-automation>
- <https://mpemagazine.co.uk/2025/08/01/leuze-on-iso-138552024-new-guidelines-for-safer-sensor-positioning-in-industrial-automation/>
- <https://www.gt-engineering.it/en/technical-standards/en-iso-standards/en-iso-13855/5-general-equations-for-the-calculation-of-the-overall-system-stopping-performance-and-minimum-distances/>

Safety program — awaiting approval of the safety functions

NOT APPROVED — the safety program is generated only after approval of the hazards, the safety functions (PLr) and their design (architecture, components, PL calculation). The standard program continues to read E-stop and interlocking only as status signals (enable, acknowledgement).

Project: Pressing and marking line LL-03 (complex example) · **Target:** Siemens F-program · **Logic:** Siemens SIMATIC S7-1200F CPU 1214FC DC/DC/DC

Items still to be approved (12)

Item	What is approved	Status
SF5 — Light curtain (AOPD) – danger point protection (finger/hand) — S4	PLr e (S2/F2/P2); stop category 0; triggered by: S4	not approved
SF5 — architecture, components and PL calculation	cat. 4, 2 channel(s), EDM yes; achieved PL — < PLr e; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC; 1 open items	not approved
SF6 — Two-hand control — S5, S6	PLr e (S2/F2/P2); stop category 0; triggered by: S5, S6	not approved
SF6 — architecture, components and PL calculation	cat. 4, 2 channel(s), EDM yes; achieved PL e ≥ PLr e; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC	not approved
SF7 — Prevention of unexpected start-up (isolation and energy dissipation, LOTO, start interlock)	PLr d (S2/F1/P2); stop category 0; triggered by: —	not approved
SF7 — architecture, components and PL calculation	cat. 3, 2 channel(s), EDM yes; achieved PL d ≥ PLr d; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC	not approved
SF8 — Safe pneumatic exhaust (SDE) and blocking of pneumatic movements (SSC)	PLr e (S2/F2/P2); stop category 0; triggered by: —	not approved
SF8 — architecture, components and PL calculation	cat. 4, 2 channel(s), EDM yes; achieved PL e ≥ PLr e; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC	not approved
SF9 — Safe stop / isolation of hydraulics (redundant valves with position monitoring)	PLr e (S2/F2/P2); stop category 0; triggered by: —	not approved
SF9 — architecture, components and PL calculation	cat. 4, 2 channel(s), EDM yes; achieved PL e ≥ PLr e; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC	not approved
SF10 — Pressure limitation (relief valve) and pressure monitoring	no PLr (mechanical principle); stop category 0	not approved
SF10 — architecture, components and PL calculation	cat. B, 1 channel(s), EDM yes; achieved PL — < PLr —; logic SIMATIC S7-1200F CPU 1214FC DC/DC/DC; 2 open items	not approved

The function design and calculations are in 13_bezpecnostni_funkce.md, the validation plan in 14_plan_validace.md, approvals in 11_schvaleni.md.

Commissioning report

NOT APPROVED — 2 of 2 items lack approval: Commissioning plan, Commissioning closure. Draft for review — do not use as a basis without approval (see 11_schvaleni.md).

Project: Pressing and marking line LL-03 (complex example) · **Created:** 03/10/2026 · **Commissioned by:** · **For the customer:**

The commissioning plan was proposed by PLCdesk from the project. For each step record the result OK / NOK / N/A, who recorded it and when, and the measured value. Commissioning is closed when all steps are OK or N/A and a responsible person approves the closure (see 11_schvaleni.md).

Phase 5 tests the E-stop and interlocks only as signals of the standard program. Safety functions are validated in phase 10.

Status: 0 OK · 0 NOK · 0 N/A · 221 without result — 221 steps in total

1. Control cabinet and power supply

ID	Step	Procedure	Expected result	Signals	Result	By / when	Measured / note
p1:visual	Visual inspection of the control cabinet	Check components and wiring against the drawings (modules DI1, DI2, DO1, AI1, AO1, terminal strips X1...X5), terminal tightness, wire marking and enclosure protection.	Components and wiring match the documentation; deviations recorded in the as-built documents.		☐		
p1:pe	Protective bonding and insulation	With the power off, measure protective bonding and insulation resistance according to the regulations for electrical equipment.	Values within limits, recorded in the inspection report.		☐		
p1:24v	24 V DC supply	Switch on the supply, measure the 24 V voltage and polarity on the distribution and the I/O modules, check the circuit protection.	24 V ±10 % at all loads, protection matches the drawing.		☐		
p1:plc	PLC and I/O modules	Switch on the PLC, download the HW configuration and the program (platforms: siemens).	CPU in RUN without errors, modules DI1, DI2, DO1, AI1, AO1 without diagnostic errors.		☐		
p1:backup	Initial backup	Before commissioning, save a backup of the program and the project.	Dated backup saved.		☐		

2. I/O loop test

ID	Step	Procedure	Expected result	Signals	Result	By / when	Measured / note
io:M1.fbkRunning	M1_fbkRunning (%I0.0) — Infeed belt conveyor – running	Without starting the drive, close the auxiliary contact of the contactor / the running signal of the drive M1 (manually on the unloaded contactor or with a jumper on the terminal).	The PLC reads M1_fbkRunning = TRUE only while closed.	M1_fbkRunning	☐		
io:M1.fault	M1_fault (%I0.1) — Infeed belt conveyor – fault	Trip the motor breaker / trigger a drive fault of M1.	The PLC reads M1_fault = TRUE on a fault and FALSE after resetting the breaker.	M1_fault	☐		
io:M2.fbkRunning	M2_fbkRunning (%I0.2) — Outfeed belt conveyor – running	Without starting the drive, close the auxiliary contact of the contactor / the running signal of the drive M2 (manually on the unloaded contactor or with a jumper on the terminal).	The PLC reads M2_fbkRunning = TRUE only while closed.	M2_fbkRunning	☐		
io:M2.fault	M2_fault (%I0.3) — Outfeed belt conveyor – fault	Trip the motor breaker / trigger a drive fault of M2.	The PLC reads M2_fault = TRUE on a fault and FALSE after resetting the breaker.	M2_fault	☐		
io:M4.fbkRunning	M4_fbkRunning (%I0.4) — Hydraulic power unit pump – running	Without starting the drive, close the auxiliary contact of the contactor / the running signal of the drive M4 (manually on the unloaded contactor or with a jumper on the terminal).	The PLC reads M4_fbkRunning = TRUE only while closed.	M4_fbkRunning	☐		

ID	Step	Procedure	Expected result	Signals	Result	By / when	Measured / note
io:M4.fault	M4_fault (%I0.5) — Hydraulic power unit pump – fault	Trip the motor breaker / trigger a drive fault of M4.	The PLC reads M4_fault = TRUE on a fault and FALSE after resetting the breaker.	M4_fault	☐		
io:Y1.fbkOpen	Y1_fbkOpen (%I0.6) — Press cylinder – open	Move cylinder Y1 manually (manual override on the valve, area secured) to the position open.	The PLC reads Y1_fbkOpen = TRUE only in the end position open.	Y1_fbkOpen	☐		
io:Y1.fbkClosed	Y1_fbkClosed (%I0.7) — Press cylinder – closed	Move cylinder Y1 manually (manual override on the valve, area secured) to the position closed.	The PLC reads Y1_fbkClosed = TRUE only in the end position closed.	Y1_fbkClosed	☐		
io:Y2.fbkOpen	Y2_fbkOpen (%I1.0) — Left clamping cylinder – open	Move cylinder Y2 manually (manual override on the valve, area secured) to the position open.	The PLC reads Y2_fbkOpen = TRUE only in the end position open.	Y2_fbkOpen	☐		
io:Y2.fbkClosed	Y2_fbkClosed (%I1.1) — Left clamping cylinder – closed	Move cylinder Y2 manually (manual override on the valve, area secured) to the position closed.	The PLC reads Y2_fbkClosed = TRUE only in the end position closed.	Y2_fbkClosed	☐		
io:Y3.fbkOpen	Y3_fbkOpen (%I1.2) — Right clamping cylinder – open	Move cylinder Y3 manually (manual override on the valve, area secured) to the position open.	The PLC reads Y3_fbkOpen = TRUE only in the end position open.	Y3_fbkOpen	☐		
io:Y3.fbkClosed	Y3_fbkClosed (%I1.3) — Right clamping cylinder – closed	Move cylinder Y3 manually (manual override on the valve, area secured) to the position closed.	The PLC reads Y3_fbkClosed = TRUE only in the end position closed.	Y3_fbkClosed	☐		
io:Y4.fbkOpen	Y4_fbkOpen (%I1.4) — Manipulator — transfer into the press – open	Move cylinder Y4 manually (manual override on the valve, area secured) to the position open.	The PLC reads Y4_fbkOpen = TRUE only in the end position open.	Y4_fbkOpen	☐		
io:Y4.fbkClosed	Y4_fbkClosed (%I1.5) — Manipulator — transfer into the press – closed	Move cylinder Y4 manually (manual override on the valve, area secured) to the position closed.	The PLC reads Y4_fbkClosed = TRUE only in the end position closed.	Y4_fbkClosed	☐		
io:Y5.fbkOpen	Y5_fbkOpen (%I1.6) — Manipulator vacuum suction cup – open	Move cylinder Y5 manually (manual override on the valve, area secured) to the position open.	The PLC reads Y5_fbkOpen = TRUE only in the end position open.	Y5_fbkOpen	☐		
io:S1.in	S1_in (%I1.7) — Emergency stop (NC)	Press and release the emergency stop button S1.	The PLC reads S1_in = TRUE at rest (normally closed contact) and FALSE when activated.	S1_in	☐		
io:S2.in	S2_in (%I2.0) — Left guard closed (NC)	Open / interrupt Left guard closed (NC) and return it to the operating position.	The PLC reads S2_in = TRUE at rest (normally closed contact) and FALSE when activated.	S2_in	☐		
io:S3.in	S3_in (%I2.1) — Right guard closed (NC)	Open / interrupt Right guard closed (NC) and return it to the operating position.	The PLC reads S3_in = TRUE at rest (normally closed contact) and FALSE when activated.	S3_in	☐		
io:S4.in	S4_in (%I2.2) — Light curtain clear	Open / interrupt Light curtain clear and return it to the operating position.	The PLC reads S4_in = FALSE at rest and TRUE when activated (input LED on the module lit).	S4_in	☐		
io:S5.in	S5_in (%I2.3) — Two-hand start — button A	Activate the signal on the machine (sensor, button, contact): Two-hand start — button A.	The PLC reads S5_in = FALSE at rest and TRUE when activated (input LED on the module lit).	S5_in	☐		
io:S6.in	S6_in (%I2.4) — Two-hand start — button B	Activate the signal on the machine (sensor, button, contact): Two-hand start — button B.	The PLC reads S6_in = FALSE at rest and TRUE when activated (input LED on the module lit).	S6_in	☐		
io:S7.in	S7_in (%I2.5) — Part in position under the manipulator	Activate the signal on the machine (sensor, button, contact): Part in position under the manipulator.	The PLC reads S7_in = FALSE at rest and TRUE when activated (input LED on the module lit).	S7_in	☐		
io:S8.in	S8_in (%I2.6) — Parts magazine empty	Activate the signal on the machine (sensor, button, contact): Parts magazine empty.	The PLC reads S8_in = FALSE at rest and TRUE when activated (input LED on the module lit).	S8_in	☐		
io:M1.outRun	M1_outRun (%Q0.0) — Infeed belt conveyor – run command	Force output M1_outRun from the PLC (watch table / force) with a short pulse; mechanics decoupled or secured.	The contactor / drive command M1 switches on; when the force is removed it drops out.	M1_outRun	☐		

ID	Step	Procedure	Expected result	Signals	Result	By / when	Measured / note
io:M2.outRun	M2_outRun (%Q0.1) — Outfeed belt conveyor – run command	Force output M2_outRun from the PLC (watch table / force) with a short pulse; mechanics decoupled or secured.	The contactor / drive command M2 switches on; when the force is removed it drops out.	M2_outRun	☐		
io:M3.outRun	M3_outRun (%Q0.2) — Vibratory parts feeder – run command	Force output M3_outRun from the PLC (watch table / force) with a short pulse; mechanics decoupled or secured.	The contactor / drive command M3 switches on; when the force is removed it drops out.	M3_outRun	☐		
io:M4.outRun	M4_outRun (%Q0.3) — Hydraulic power unit pump – run command	Force output M4_outRun from the PLC (watch table / force) with a short pulse; mechanics decoupled or secured.	The contactor / drive command M4 switches on; when the force is removed it drops out.	M4_outRun	☐		
io:Y1.outOpen	Y1_outOpen (%Q0.4) — Press cylinder – open command	Force output Y1_outOpen from the PLC (watch table / force) with a short pulse; mechanics decoupled or secured.	The coil of valve Y1 energizes (LED on the connector) and the cylinder moves to the open position; when released it returns.	Y1_outOpen	☐		
io:Y2.outOpen	Y2_outOpen (%Q0.5) — Left clamping cylinder – open command	Force output Y2_outOpen from the PLC (watch table / force) with a short pulse; mechanics decoupled or secured.	The coil of valve Y2 energizes (LED on the connector) and the cylinder moves to the open position; when released it returns.	Y2_outOpen	☐		
io:Y3.outOpen	Y3_outOpen (%Q0.6) — Right clamping cylinder – open command	Force output Y3_outOpen from the PLC (watch table / force) with a short pulse; mechanics decoupled or secured.	The coil of valve Y3 energizes (LED on the connector) and the cylinder moves to the open position; when released it returns.	Y3_outOpen	☐		
io:Y4.outOpen	Y4_outOpen (%Q0.7) — Manipulator — transfer into the press – open command	Force output Y4_outOpen from the PLC (watch table / force) with a short pulse; mechanics decoupled or secured.	The coil of valve Y4 energizes (LED on the connector) and the cylinder moves to the open position; when released it returns.	Y4_outOpen	☐		
io:Y5.outOpen	Y5_outOpen (%Q1.0) — Manipulator vacuum suction cup – open command	Force output Y5_outOpen from the PLC (watch table / force) with a short pulse; mechanics decoupled or secured.	The coil of valve Y5 energizes (LED on the connector) and the cylinder moves to the open position; when released it returns.	Y5_outOpen	☐		
io:Y6.outOpen	Y6_outOpen (%Q1.1) — Marking unit (stamping cylinder) – open command	Force output Y6_outOpen from the PLC (watch table / force) with a short pulse; mechanics decoupled or secured.	The coil of valve Y6 energizes (LED on the connector) and the cylinder moves to the open position; when released it returns.	Y6_outOpen	☐		
io:H1.out	H1_out (%Q1.2) — Beacon green — running	Force output H1_out from the PLC (watch table / force) with a short pulse; mechanics decoupled or secured.	Beacon green — running switches on; when the force is removed it switches off.	H1_out	☐		
io:H2.out	H2_out (%Q1.3) — Beacon red — fault	Force output H2_out from the PLC (watch table / force) with a short pulse; mechanics decoupled or secured.	Beacon red — fault switches on; when the force is removed it switches off.	H2_out	☐		
io:H3.out	H3_out (%Q1.4) — Horn — cycle start	Force output H3_out from the PLC (watch table / force) with a short pulse; mechanics decoupled or secured.	Horn — cycle start switches on; when the force is removed it switches off.	H3_out	☐		
io:H4.out	H4_out (%Q1.5) — Electromagnetic guard lock	Force output H4_out from the PLC (watch table / force) with a short pulse; mechanics decoupled or secured.	Electromagnetic guard lock switches on; when the force is removed it switches off.	H4_out	☐		
io:B1.raw	B1_raw (%IW64) — Hydraulic pressure – bar	Connect a calibrator instead of the sensor and set 0 %, 50 % and 100 % of the signal range.	The PLC shows 0, 125 and 250 bar (raw value 0 / 13824 / 27648).	B1_raw	☐		
io:B2.raw	B2_raw (%IW66) — Oil temperature – °C	Connect a calibrator instead of the sensor and set 0 %, 50 % and 100 % of the signal range.	The PLC shows 0, 50 and 100 °C (raw value 0 / 13824 / 27648).	B2_raw	☐		

ID	Step	Procedure	Expected result	Signals	Result	By / when	Measured / note
io:B3.raw	B3_raw (%IW68) — Press force (strain gauge) – kN	Connect a calibrator instead of the sensor and set 0 %, 50 % and 100 % of the signal range.	The PLC shows 0, 200 and 400 kN (raw value 0 / 13824 / 27648).	B3_raw	☐		
io:B4.raw	B4_raw (%IW70) — Press position (linear sensor) – mm	Connect a calibrator instead of the sensor and set 0 %, 50 % and 100 % of the signal range.	The PLC shows 0, 250 and 500 mm (raw value 0 / 13824 / 27648).	B4_raw	☐		
io:B5.raw	B5_raw (%IW72) — Cooling flow – l/min	Connect a calibrator instead of the sensor and set 0 %, 50 % and 100 % of the signal range.	The PLC shows 0, 30 and 60 l/min (raw value 0 / 13824 / 27648).	B5_raw	☐		
io:U1.raw	U1_raw (%QW64) — Proportional pressure valve – %	Write 0 %, 50 % and 100 % of the range from the PLC (0 / 50 / 100 %).	0 / 50 / 100 % of the signal measured at the terminal; the actuator responds in the corresponding direction.	U1_raw	☐		
io:U2.raw	U2_raw (%QW66) — Conveyor M2 speed setpoint – %	Write 0 %, 50 % and 100 % of the range from the PLC (0 / 50 / 100 %).	0 / 50 / 100 % of the signal measured at the terminal; the actuator responds in the corresponding direction.	U2_raw	☐		

3. Drives

ID	Step	Procedure	Expected result	Signals	Result	By / when	Measured / note
drv:M1:dir	M1: direction of rotation	In manual mode briefly start M1 (manRun_M1) with the mechanics decoupled or secured.	Direction matches the process; if reversed, swap the phases / change the drive parameter.	manRun_M1, M1_outRun	☐		
drv:M1:fbk	M1: start-up and running feedback	Start M1 manually and measure the time from command M1_outRun to feedback M1_fbkRunning; then stop it and check that the feedback drops out.	Running feedback within 2.4 s (block timeout 3 s, 20 % margin). Record the measured time.	M1_outRun, M1_fbkRunning	☐		
drv:M1:fault	M1: drive fault	While running manually, trip the breaker / drive fault of M1.	Block M1 faulted (alarm A_M1_FAULT), output off; after resetting the breaker and acknowledging (cmdAck) it can be started again.	M1_fault, cmdAck	☐		
drv:M2:dir	M2: direction of rotation	In manual mode briefly start M2 (manRun_M2) with the mechanics decoupled or secured.	Direction matches the process; if reversed, swap the phases / change the drive parameter.	manRun_M2, M2_outRun	☐		
drv:M2:fbk	M2: start-up and running feedback	Start M2 manually and measure the time from command M2_outRun to feedback M2_fbkRunning; then stop it and check that the feedback drops out.	Running feedback within 2.4 s (block timeout 3 s, 20 % margin). Record the measured time.	M2_outRun, M2_fbkRunning	☐		
drv:M2:fault	M2: drive fault	While running manually, trip the breaker / drive fault of M2.	Block M2 faulted (alarm A_M2_FAULT), output off; after resetting the breaker and acknowledging (cmdAck) it can be started again.	M2_fault, cmdAck	☐		
drv:M3:dir	M3: direction of rotation	In manual mode briefly start M3 (manRun_M3) with the mechanics decoupled or secured.	Direction matches the process; if reversed, swap the phases / change the drive parameter.	manRun_M3, M3_outRun	☐		
drv:M4:dir	M4: direction of rotation	In manual mode briefly start M4 (manRun_M4) with the mechanics decoupled or secured.	Direction matches the process; if reversed, swap the phases / change the drive parameter.	manRun_M4, M4_outRun	☐		
drv:M4:fbk	M4: start-up and running feedback	Start M4 manually and measure the time from command M4_outRun to feedback M4_fbkRunning; then stop it and check that the feedback drops out.	Running feedback within 2.4 s (block timeout 3 s, 20 % margin). Record the measured time.	M4_outRun, M4_fbkRunning	☐		
drv:M4:fault	M4: drive fault	While running manually, trip the breaker / drive fault of M4.	Block M4 faulted (alarm A_M4_FAULT), output off; after resetting the breaker and acknowledging (cmdAck) it can be started again.	M4_fault, cmdAck	☐		

ID	Step	Procedure	Expected result	Signals	Result	By / when	Measured / note
drv:Y1:travel	Y1: travel and end positions	In manual mode open and close Y1 (manOpen_Y1) and measure the travel time in both directions.	End position within 4 s (block timeout 5 s, 20 % margin); limit switches match the real position. Record the measured times.	manOpen_Y1, Y1_outOpen, Y1_fbkOpen, Y1_fbkClosed	☐		
drv:Y1:hold	Y1: held position monitoring	In the open position briefly interrupt limit switch Y1_fbkOpen (unplug the sensor connector).	Block fault (alarm A_Y1_POS), output off; after restoring and acknowledging it can be operated again.	Y1_fbkOpen, cmdAck	☐		
drv:Y2:travel	Y2: travel and end positions	In manual mode open and close Y2 (manOpen_Y2) and measure the travel time in both directions.	End position within 3.2 s (watchdog time of step 6 is 4 s, 20 % margin; block timeout 5 s); limit switches match the real position. Record the measured times.	manOpen_Y2, Y2_outOpen, Y2_fbkOpen, Y2_fbkClosed	☐		
drv:Y2:hold	Y2: held position monitoring	In the open position briefly interrupt limit switch Y2_fbkOpen (unplug the sensor connector).	Block fault (alarm A_Y2_POS), output off; after restoring and acknowledging it can be operated again.	Y2_fbkOpen, cmdAck	☐		
drv:Y3:travel	Y3: travel and end positions	In manual mode open and close Y3 (manOpen_Y3) and measure the travel time in both directions.	End position within 3.2 s (watchdog time of step 7 is 4 s, 20 % margin; block timeout 5 s); limit switches match the real position. Record the measured times.	manOpen_Y3, Y3_outOpen, Y3_fbkOpen, Y3_fbkClosed	☐		
drv:Y3:hold	Y3: held position monitoring	In the open position briefly interrupt limit switch Y3_fbkOpen (unplug the sensor connector).	Block fault (alarm A_Y3_POS), output off; after restoring and acknowledging it can be operated again.	Y3_fbkOpen, cmdAck	☐		
drv:Y4:travel	Y4: travel and end positions	In manual mode open and close Y4 (manOpen_Y4) and measure the travel time in both directions.	End position within 4 s (block timeout 5 s, 20 % margin); limit switches match the real position. Record the measured times.	manOpen_Y4, Y4_outOpen, Y4_fbkOpen, Y4_fbkClosed	☐		
drv:Y4:hold	Y4: held position monitoring	In the open position briefly interrupt limit switch Y4_fbkOpen (unplug the sensor connector).	Block fault (alarm A_Y4_POS), output off; after restoring and acknowledging it can be operated again.	Y4_fbkOpen, cmdAck	☐		
drv:Y5:travel	Y5: travel and end positions	In manual mode open and close Y5 (manOpen_Y5) and measure the travel time in both directions.	End position within 1.6 s (watchdog time of step 4 is 2 s, 20 % margin; block timeout 5 s); limit switches match the real position. Record the measured times.	manOpen_Y5, Y5_outOpen, Y5_fbkOpen	☐		
drv:Y5:hold	Y5: held position monitoring	In the open position briefly interrupt limit switch Y5_fbkOpen (unplug the sensor connector).	Block fault (alarm A_Y5_POS), output off; after restoring and acknowledging it can be operated again.	Y5_fbkOpen, cmdAck	☐		
drv:Y6:travel	Y6: travel and end positions	In manual mode open and close Y6 (manOpen_Y6) and measure the travel time in both directions.	End position within 4 s (block timeout 5 s, 20 % margin); limit switches match the real position. Record the measured times.	manOpen_Y6, Y6_outOpen	☐		

Excerpt: first phases of the commissioning plan. The full record contains all phases and is also available as CSV for printing.

```
// Gen_Main.scl - machine block (multi-instance). Import AFTER Gen_Library.scl.

FUNCTION_BLOCK "FB_Machine"
{ S7_Optimized_Access := 'TRUE' }
VERSION : 0.2
VAR_INPUT
    enableIn : Bool;    // central enable from outside (OB1)
END_VAR
VAR
    enable : Bool;
    instM1 : "FB_Motor";
    instM2 : "FB_Motor";
    instM3 : "FB_Motor";
    instM4 : "FB_Motor";
    instY1 : "FB_Ventil";
    instY2 : "FB_Ventil";
    instY3 : "FB_Ventil";
    instY4 : "FB_Ventil";
    instY5 : "FB_Ventil";
    instY6 : "FB_Ventil";
    instB1 : "FB_AnalogIn";
    instB2 : "FB_AnalogIn";
    instB3 : "FB_AnalogIn";
    instB4 : "FB_AnalogIn";
    instB5 : "FB_AnalogIn";
    instU1 : "FB_AnalogOut";
    instU2 : "FB_AnalogOut";
    modeAuto : BOOL;    // TODO: mode selector (HMI); FALSE = manual mode
    cmdAutoStart : BOOL; // TODO: auto start button
    cmdAck : BOOL;      // TODO: fault acknowledge button (HMI)
    machineFault : BOOL; // machine fault (block error / step timeout); latched until acknowledged
    manRun_M1 : BOOL;   // TODO: manual command from HMI (effective when AUTO is off)
    manRun_M2 : BOOL;   // TODO: manual command from HMI (effective when AUTO is off)
    manRun_M3 : BOOL;   // TODO: manual command from HMI (effective when AUTO is off)
    manRun_M4 : BOOL;   // TODO: manual command from HMI (effective when AUTO is off)
    manOpen_Y1 : BOOL;  // TODO: manual command from HMI (effective when AUTO is off)
    manOpen_Y2 : BOOL;  // TODO: manual command from HMI (effective when AUTO is off)
    manOpen_Y3 : BOOL;  // TODO: manual command from HMI (effective when AUTO is off)
    manOpen_Y4 : BOOL;  // TODO: manual command from HMI (effective when AUTO is off)
    manOpen_Y5 : BOOL;  // TODO: manual command from HMI (effective when AUTO is off)
    manOpen_Y6 : BOOL;  // TODO: manual command from HMI (effective when AUTO is off)
    faultStep : INT;    // sequence step in which the time expired (diagnostics)
    seqStep : INT;
    seqRun_M3 : BOOL;
    seqRun_M1 : BOOL;
    seqOpen_Y4 : BOOL;
    seqOpen_Y5 : BOOL;
    seqOpen_Y2 : BOOL;
    seqOpen_Y3 : BOOL;
    seqRun_M4 : BOOL;
    seqOpen_Y1 : BOOL;
    seqOpen_Y6 : BOOL;
    seqRun_M2 : BOOL;
    tonSeq10 : TON_TIME;
    tonSeq20 : TON_TIME;
    tonSeq30 : TON_TIME;
    tonSeq40 : TON_TIME;
    tonSeq50 : TON_TIME;
    tonSeq60 : TON_TIME;
    tonSeq70 : TON_TIME;
    tonSeq80 : TON_TIME;
    tonSeq90 : TON_TIME;
    tonSeq100 : TON_TIME;
    tonSeq110 : TON_TIME;
    tonSeq120 : TON_TIME;
    tonSeq130 : TON_TIME;
    tonSeq140 : TON_TIME;
    tonSeq150 : TON_TIME;
    tonSeq160 : TON_TIME;
    tonSeq170 : TON_TIME;
    tonSeq180 : TON_TIME;
END_VAR

BEGIN
    #enable := #enableIn;

    // --- Automatic sequence (AUTO mode) ---
    // AUTO off, loss of enable or machine fault: sequence to step 0, commands off
    IF NOT #modeAuto OR NOT #enable OR #machineFault THEN
        #seqStep := 0;
        #seqRun_M3 := FALSE;
        #seqRun_M1 := FALSE;
        #seqOpen_Y4 := FALSE;
    
```

```

#seqOpen_Y5 := FALSE;
#seqOpen_Y2 := FALSE;
#seqOpen_Y3 := FALSE;
#seqRun_M4 := FALSE;
#seqOpen_Y1 := FALSE;
#seqOpen_Y6 := FALSE;
#seqRun_M2 := FALSE;
END_IF;

CASE #seqStep OF
  0: // waiting for start
    IF #modeAuto AND #enable AND NOT #machineFault AND #cmdAutoStart THEN #seqStep := 10; END_IF;
  10: // Step 1: M3 start
    #seqRun_M3 := TRUE;
    IF #tonSeq10.Q THEN #seqStep := 20; END_IF;
  20: // Step 2: M1 start
    #seqRun_M1 := TRUE;
    IF "M1_fbkRunning" THEN #seqStep := 30;
    ELSIF #tonSeq20.Q THEN #machineFault := TRUE; #faultStep := 20; // step timeout 3 s
    END_IF;
  30: // Step 3: Y4 open
    #seqOpen_Y4 := TRUE;
    IF "Y4_fbkOpen" THEN #seqStep := 40;
    ELSIF #tonSeq30.Q THEN #machineFault := TRUE; #faultStep := 30; // step timeout 5 s
    END_IF;
  40: // Step 4: Y5 open
    #seqOpen_Y5 := TRUE;

```

(* ... excerpt, the full program is in the project file set *)